The Generalized-Sturmian Secular Equation Block-Encodes Without a Metric: A Quantum-Algorithm Analysis, Metric-Free for Atoms and a Conditioning Frontier for Molecules*

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(Dated: August 18, 2026)

Central result. In Avery's isoenergetic (potential-weighted) posing, the *atomic* Coulomb–Sturmian secular equation is a standard, metric-free eigenproblem—an apparently novel quantum-encoding target, with both documented cost-risks of the genre (metric conditioning; an outer energy search) structurally absent—whereas the naive L^2 posing of the same basis inflates the cost instead. Its block-encoding 1-norm grows *more slowly than the configuration count* over every basis we can compute, and *more* slowly as the basis grows. That advantage, however, belongs to the *basis-growth rule* rather than to the construction: it holds when $l_{\max}$ is fixed and higher-n configurations are added, and weakens when the basis is grown as full hydrogenic shells — there the total exponent rises toward 1 ($0.867 \to 0.911$) while remaining below it, and only the pure-number block T' itself goes superlinear. The cheap rule also carries an accuracy floor, whose mechanism we locate exactly: the metric-free form holds *iff* the basis scale is locked to the eigenvalue, $\lambda = p_\kappa = \sqrt{-2E}$, and that lock is not the variational optimum. Solving the *same* span variationally with the scale freed reaches chemical accuracy, but hands back *both* of the cost risks named above — the L^2 metric returns, and the scale becomes an outer search — along with the encoding advantage itself (1-norm growth from $K^{0.72}$ to $K^{2.75}$ on the s-only comparison ladder, $\mathrm{cond}(S) \sim K^{0.94}$). The absence of those two risks and the presence of the floor are therefore not independent virtues and defects but one structural fact. The floor is moreover a *ground-state* pathology rather than a property of the method: because $\|M\|_1$ does not depend on which root is extracted, every ^1S state costs the same to encode, and at the largest basis where both roots were computed, $K = 452$, the ground state sits 4.28× above chemical accuracy while $2\,^1S$ sits 1.08× above it; and on the s-only ladder at $K = 105$ the posing cost falls by 4.3×, 3.0× and 2.6× across the first three rungs of the ^1S ladder. The result is therefore a measured trade-off between cost growth, attainable accuracy and *which state is targeted*, not an unconditional efficiency claim. The development follows.

A recurring intuition in quantum simulation is that a Coulomb–Sturmian / hyperspherical basis is well suited to a quantum computer, because its angular matrix elements are exact rationals generated on the device from quantum-number labels rather than loaded as a classically-computed tensor; only the one-dimensional radial content need be shipped in. We test the load-bearing obstruction to that idea—the cost of *encoding* the basis—and find it turns sharply on how the basis is posed. **[MEASURED]** Encoded naively, in the ordinary L^2 Schrödinger framework, the genuine shared-scale Coulomb–Sturmian basis is L^2-non-orthogonal; a second-quantized (Jordan–Wigner) encoding must Löwdin-orthogonalize it, and the LCU 1-norm λ then *inflates* as $\lambda \sim Q^{3.3}$ (versus $Q^{1.2}$ for an orthonormal hydrogenic basis), driven by the dense $S^{-1/2}$ that Löwdin orthogonalization introduces as the basis members grow progressively linearly dependent. **[ESTABLISHED, from Avery's formulation]** The generalized-Sturmian *isoenergetic* reformulation removes this obstruction for atoms: posed in the potential-weighted metric in which the Coulomb–Sturmians are orthonormal, the atomic secular equation is a *standard* (metric-free) eigenproblem $[\,\mathrm{diag}(ZR_\nu) + T' - p_\kappa \mathbb{1}\,]B = 0$ whose eigenvalues are the scaling parameters $p_\kappa = \sqrt{-2E}$ directly and whose interelectron matrix T' is a matrix of pure numbers, independent of p_κ, of energy, and of Z. **[MEASURED]** We implement it, validate it against helium (a single Goscinskian configuration reproduces the textbook variational value −2.847 Ha; adding configurations lowers the energy monotonically through the s, $+p$ and *spdf* sectors—past the locked s-sector value −2.8745 Ha to −2.8964 Ha, toward the exact non-relativistic −2.90372 Ha), and measure the decisive quantity: the block-encoding 1-norm of the secular matrix grows *more slowly than the matrix dimension*, $\|M\|_1 \sim K^{0.82}$ over the computed range $K = 74$–164 (full $s{+}p{+}d{+}f$, helium; $K^{0.73}$ s-only)—the opposite of the naive inflation. The exponent is a fit over a window and not an asymptotic regime: its local slope *falls* monotonically with K (Sec. IV), it moves with the nuclear charge, and it belongs to the fixed-$l_{\max}$ growth rule rather than to the construction. **[OBSERVATION]** A targeted prior-art search finds no existing quantum algorithm combining a Sturmian basis, a quantum eigenvalue routine, and the isoenergetic inversion; the two documented cost risks for such an algorithm—metric conditioning and an outer energy search—are both *absent* in the atomic case (no metric; the eigenvalues are the energies). **[MEASURED]** For molecules the Shibuya–Wulfman metric returns and the problem is again generalized; we measure its conditioning and find it uniformly better than the L^2 overlap—its intra-center block is exactly the identity—and markedly better at larger internuclear separation, but still growing with basis size through the inter-center coupling. **[PRIOR ART]** The growth law is not new: the band-limited concentration rate $1 - \sigma_{\max} \propto (kR)^2/n^2$, asymptotic exponent

2, is the Kac–Murdock–Szegő asymptotic, and what is ours is the *identification* of the Shibuya–Wulfman metric as such a finite section (the fitted window exponents 1.85/1.97 are pre-asymptotic readings). **[MEASURED]** The same cross-center singular spectrum carries the composition-wall commutator, whose norm saturates at $\frac{1}{2}$. **[RESOURCE MODEL]** Turning this into a block-encoding count for H_2^+ (validated to bind, 1.3% s-only), the payoff is the gerade/ungerade lever: the σ_g ground state lives in the gerade sector, whose conditioning is *flat*—and flat at an exact constant of the sinc symbol, $2/(1+\min_x j_0(x)) = 2.555041\ldots$, independent of R and of basis size—so the ground-state metric penalty does *not* grow with basis size—a $\sim$60$\times$ reduction versus the full metric and 10^3–$10^4\times$ smaller than a standard Gaussian metric for the same problem (a metric-versus-metric ratio that scales with the Gaussian exponent spread, hence illustrative rather than a fixed factor), at $\sim$30 qubits. **[MEASURED]** The penalty also does not compound with electron number: the metric is discharged once at the one-electron molecular-orbital construction, after which the many-electron secular equation carries an identity metric—the metric cost is one-electron, not N-electron. **[MEASURED]** A third lever removes the conditioning cost outright where it applies: the ill-conditioning is a symbol zero of known order and location, so a band-Toeplitz preconditioner bounds cond flat in basis size, and—unlike the gerade lever—it does not require equivalent centers, reaching water's A_1 block (bounded at 44 out to $N = 192$, against a raw $N^{1.97}$), provided the band is aligned to the symbol's null direction: in the unrotated frame the same band is 106$\times$ *worse* than no treatment at all. **[RESOURCE MODEL]** A direct block-encoding of the preconditioned metric then takes the penalty from n^3 to n (the circulant-embedded circuit is analysed, not compiled). **[OPEN]** The atomic case is a clean, apparently novel, metric-free quantum secular equation; the molecular case is a frontier on which the *conditioning* axis is removable on s-sector shared-scale bases at $M = 2$ and non-collinear $M = 3$, the *locality* cost is capped by the symbol's other pole, and the ℓ-block sparsity loss is structural and not a conditioning effect at all. **[MEASURED]** Its remaining question—whether the N-electron 1-norm stays sublinear as it does for atoms—we answer in the negative: the atomic sublinearity rides on a diagonal nuclear matrix $T^0 = ZR_\nu$ that is single-center-specific, and a validated two-center CI (dissociating H_2 correctly) has a *standard* block-encoding 1-norm growing polynomially ($\sim n_{\mathrm{orb}}^{2.2}$), with no sublinear behavior. The real molecular savings are the metric levers (the band-Toeplitz preconditioner, the gerade sector, large separation, one-electron-only metric), not a sublinear matrix. We do not demonstrate the algorithm on hardware; this is a structural and resource analysis.

I. INTRODUCTION

The generalized-Sturmian method of Avery and Avery [1–4], built on Goscinski's [39] weighted wave equation and the momentum-space Fock projection, is one of the most elegant classical frameworks for atomic and molecular electronic structure. Its angular content is closed-form—Gaunt integrals, Wigner $3j/6j$ symbols, Shibuya–Wulfman integrals—generated from the quantum numbers (n, ℓ, m) with no spatial quadrature. This is the same discrete angular skeleton that the Geometric Vacuum program reads as a graph (Papers [15, 16]); the two frameworks meet exactly at those closed-form angular integrals.

The angular cleanness suggests a quantum-computing use, articulated independently in the Sturmian and Geometric-Vacuum settings: the bottleneck in quantum simulation is often not the term count of the Hamiltonian but the *input/output*—classically computing a two-electron integral tensor and loading it onto the device. If the angular structure can be generated on the device from labels, and only the one-dimensional radial factors loaded, the loading cost is dominated by a small radial-seed count rather than an $O(M^4)$ tensor. This is a known-good lever in fault-tolerant simulation: plane-wave quantum-simulation methods compute their coefficients rather than loading them and win decisively for large basis sets [7, 8] (first quantization, with $O(\eta \log N)$ qubits, at the [8] end).

The lever has a well-documented Achilles heel, however. Its payoff is governed not by the loading count alone but by the *LCU 1-norm* $\lambda = \sum_i |c_i|$ of the encoded Hamiltonian, which sets the query and T-gate cost of qubitization and block-encoding [9, 10]; plane waves win the I/O yet pay a large λ. The question this paper addresses is whether a Coulomb–Sturmian encoding keeps λ bounded. We find that the answer depends entirely on *how the basis is posed*, and that the isoenergetic reformulation—the defining move of the generalized-Sturmian method—is precisely what makes the atomic problem cheap.

The paper is organized as the argument runs. Section II measures the naive obstruction. Section III states the isoenergetic reformulation and why it is metric-free for atoms. Section IV implements and validates it on helium and measures the sublinear 1-norm. Section V states the quantum algorithm and its novelty. Section VI measures the molecular Shibuya–Wulfman conditioning. Section VII states the honest boundary.

* Paper 60 in the Geometric Vacuum series

II. THE NAIVE OBSTRUCTION: 1-NORM INFLATION FROM L^2 ORTHOGONALIZATION

A second-quantized (Jordan–Wigner) encoding requires an L^2-orthonormal set of single-particle spin-orbitals. The genuine shared-scale Coulomb–Sturmians are *not* L^2-orthonormal: at a common scale k their higher members grow linearly dependent, so a qubit encoding must first orthogonalize them (Löwdin, $S^{-1/2}$), and the coefficient distribution—hence $\lambda = \sum_i |c_i|$—spreads.

[**MEASURED**] In a controlled s-sector helium construction in which only the radial convention is changed and the ground shell is anchored so the two families coincide at $Q = 2$ qubits, the LCU 1-norm scales as

$$\lambda_{\text{hydrogenic}} \sim Q^{1.19}, \qquad \lambda_{\text{Sturmian}} \sim Q^{3.33}, \qquad (1)$$

a factor of ~ 18 larger by ten qubits (a measured ratio from the $\lambda(Q)$ data; like Eq. (6) these exponents are numerical fits, and the tracked claim is the sublinear-vs-superlinear contrast, not the precise value). The mechanism is the L^2 overlap S: with helium Coulomb–Sturmians its condition number climbs $(3.0, 5.8, 13.9, 32.2$ for $N = 2, 3, 5, 8$ — growing, but modest), and the dense $S^{-1/2}$ built from that increasingly non-orthogonal Gram inflates λ. The inflation is driven by the *density* of $S^{-1/2}$ and the spread it induces in the coefficient distribution, not by numerical instability: a condition number of 32 is not ill-conditioned. This is a currency-change of the familiar cost-conservation phenomenon in non-orthogonal quantum chemistry (the overlap metric never disappears; moving it densifies operators, states, or coefficients [15]). Encoded this way, the Sturmian basis is the *worst*, not the best, choice.

[**MEASURED**] The obstruction is intrinsic to the L^2 posing rather than to a choice of orthogonalization: the L^2 Löwdin encoding inflates the LCU 1-norm as $\lambda \sim Q^{3.33}$ against $Q^{1.19}$ for an orthonormal hydrogenic basis (Eq. (1)), measured through a faithful Jordan–Wigner LCU. That is the cost obstruction, and it stands. *Correction, 2026-09-07 — a stronger claim is withdrawn here.* Earlier versions added that re-solving the converged $s+p+d+f$ helium problem as the generalized eigenproblem $MB = pSB$ *diverges*, with the metric condition number climbing $4 \to 3673$ and the energy running to -3×10^6 Ha. Both numbers are artifacts of the fixed 60-bohr radial domain [retracted 2026-09-07: p60-l2-metric-diverges] (Appendix A). On a converged domain $\text{cond}(S) = 16.0$ at the same point, and the growth is mild and smooth, $\text{cond}(S) \approx 0.12\,K$ — 4.07 at $K = 10$, 23.5 at $K = 164$, 56.3 at $K = 452$. The blow-up switches on exactly where $n^2_{\max}$ first exceeds the domain size, and every domain agrees to four digits below that point. The L^2 Gram matrix of this basis is therefore ordinary, not ill-conditioned, and we no longer claim the metric-free posing is the *only* well-conditioned one. What we do claim is the cost statement above: the L^2 posing is expensive to *encode*, which is a different assertion from being numerically unstable, and is the one this paper's argument needs.

[**PRIOR ART**] The classical momentum-space literature met this and stepped around it, and *how* it did so prices exactly what the quantum encoding costs. Monkhorst and Jeziorski [36] titled a 1979 note "No linear dependence or many-center integral problems in momentum space quantum chemistry", and its abstract states that for a many-center *one-electron* system the eigenvalues follow from "diagonalizations of simple overlap matrices" and that "the problems of many-center integrals and instabilities due to overcompleteness of basis sets do not appear at all". Taken at face value that contradicts Eq. (13). It does not, and the reconciliation is worth stating because it locates our own cost.

The degeneracy is not absent from their construction: it is the same pencil, the same matrix of translation phases, and $\sigma_{\max} \to 1$ at $p = 0$ is a property of the symbol, indifferent to who is looking. What differs is the extraction. Letting the overlap enter *only multiplicatively* — solving a determinantal condition in the scale as a scalar root search, or diagonalising the overlap directly — inverts nothing, so a near-null direction produces a harmless spurious branch instead of amplified error; and the basis is exactly orthonormal *within each center* in the metric actually used — the statement this paper measures independently as the intra-center block being the identity, and not a statement about the two-center set, whose smallest generalized eigenvalue does collapse — so the L^2 Gram matrix that carries the classical linear-dependence problem never enters at all. **They do not remove the degeneracy; they keep it out of the denominator.** We invert because a block-encoding wants a standard Hermitian eigenproblem, and that requirement — not the basis, and not the metric — is what converts a benign spectral feature into a resource multiplier.

The trade runs both ways and we do not claim the better side of it: a determinantal re-posing removes the conditioning multiplier but reinstates the outer nonlinear search over the scale that Eq. (3) exists to eliminate, and which of the two is cheaper at scale is an open resource question this paper does not answer. *Provenance, because it bounds the paragraph:* the bibliographic record and the abstract are verified at source; the two-page body is closed with no repository copy and is unread, so the mechanism above is reconstructed from the surrounding lineage rather than read.

III. THE ISOENERGETIC REFORMULATION IS METRIC-FREE FOR ATOMS

The Coulomb–Sturmians obey a *potential-weighted* orthonormality relation [3, 5, 6],

$$\int d^3x\, \varphi^*_{n'\ell'm'}(x)\, \frac{1}{r}\, \varphi_{n\ell m}(x) = \frac{k}{n}\, \delta_{n'n}\delta_{\ell'\ell}\delta_{m'm}, \qquad (2)$$

orthonormal in the $1/r$ weight, *not* in L^2. [**MEASURED**] We confirm Eq. (2) numerically: the potential-

weighted overlap is diagonal to $\sim 10^{-6}$ off-diagonal, whereas the L^2 overlap of the same basis is the non-orthogonal matrix of Sec. II.

Building the electronic-structure problem in this native metric is the generalized-Sturmian method. With the basis potential taken to be the bare nuclear attraction $V_0 = -\sum_j Z/r_j$, the Goscinskian configurations Φ_ν are Slater determinants of hydrogenlike orbitals sharing one weighted charge $Q_\nu = p_\kappa/R_\nu$, with configuration root $R_\nu = \sqrt{\sum_j 1/n_j^2}$, all isoenergetic at $E = -\frac{1}{2}p_\kappa^2$. The kinetic term cancels against V_0 by construction, the nuclear-attraction matrix is diagonal, $T^0_{\nu'\nu} = ZR_\nu\,\delta_{\nu'\nu}$, and the atomic secular equation is [3, 4]

$$\sum_\nu \Big[\, ZR_\nu\,\delta_{\nu'\nu} + T'_{\nu'\nu} - p_\kappa\,\delta_{\nu'\nu}\,\Big] B_\nu = 0, \qquad E = -\tfrac{1}{2}p_\kappa^2. \tag{3}$$

Three features of Eq. (3) are load-bearing for a quantum algorithm, and all are consequences of the potential-weighting:

1. It is a *standard* eigenproblem, $MB = p_\kappa B$ with $M = \mathrm{diag}(ZR_\nu) + T'$. There is *no overlap metric* S: the potential-weighting has absorbed it into the diagonal nuclear term. The non-orthogonality of Sec. II is not repaired, it is *never present*.

2. Its eigenvalues are the scaling parameters p_κ *directly*; one diagonalization yields the whole spectrum via $E = -\frac{1}{2}p_\kappa^2$. There is no self-consistency and no outer search over the energy.

3. The interelectron matrix $T'_{\nu'\nu} = -p_\kappa^{-1}\int \Phi^*_{\nu'} V' \Phi_\nu$ is a matrix of *pure numbers*, independent of p_κ, of E, and of Z: the two-electron integrals scale as $Q_\nu = p_\kappa/R_\nu$, and the $-p_\kappa^{-1}$ prefactor cancels the p_κ. A single T' serves an entire isoelectronic series.

T' is assembled by Slater–Condon reduction of the two-electron operator, with the $1/r_{12}$ Laplace expansion giving a finite multipole sum (the angular factors are products of Gaunt integrals / Wigner $3j$, recoupling to $6j$ for coupled configurations) times a radial Slater integral [4].

IV. ATOMIC IMPLEMENTATION, VALIDATION, AND THE SUBLINEAR 1-NORM

We implement Eq. (3) for helium ($Z = 2$). For the leading configuration $1s^2$ the matrix is 1×1: $R_{1s^2} = \sqrt{2}$, the unit-charge integral is $\langle 1s^2|r_{12}^{-1}|1s^2\rangle = \frac{5}{8}$, so

$$T'_{1s^2,1s^2} = -\tfrac{5}{8}/\sqrt{2} \approx -0.442, \quad p_\kappa = Z\sqrt{2} + T' = 2.386, \tag{4}$$

giving $E = -p_\kappa^2/2 = -2.847$ Ha. **[MEASURED]** Computed on a radial grid this reproduces the textbook variational helium value (-2.84766 Ha; the pure number $\frac{5}{8}/\sqrt{2}$ is confirmed to grid precision), and the "bare" matrix (no V') returns -4.0 Ha, i.e. two He^+ $1s$ orbitals—the exact non-interacting limit. **[MEASURED]** Adding configurations coupled to total ^1S lowers the energy monotonically,

$$-2.8477\,(1s^2) \;\to\; -2.8745\,(s,\,K{=}55) \;\to\; -2.8947\,(+p,\,K{=}100) \;\to$$

converging toward the exact non-relativistic -2.90372 Ha. Every rung beyond the leading $1s^2$ single configuration (a 1×1 matrix) is taken at the same $n_{\max} = 10$ family, so that part of the sequence is a pure $\ell_{\max}$ ladder; values are rounded to four decimals. **[MEASURED]** The residual is *not* a deficiency of the configuration span. A variational CI over the *identical* span, with the global scale optimized, reaches 1.28 mHa of the exact energy at $K = 130$, where the isoenergetic solution of that same span stalls at 7.46 mHa; in the s-sector, where the exact s-limit -2.879029 Ha is known independently, the same comparison is 0.15 mHa against 4.40 mHa at $K = 136$. The residual is therefore the price of the *posing*, not of the basis: the metric-free form requires the basis scale to be locked to the eigenvalue, $\lambda = p_\kappa$, and that lock is not the variational optimum. **[INTERNAL THEOREM]** The same identity makes the bound automatic rather than fortunate—the isoenergetic root is the lowest root of the fixed-scale generalized problem $H(p_\kappa)C = E\,SC$, so every point lies above the exact value by construction, not by observation. **[OBSERVATION]** The canon does not appear to state this. Its variational discussion is framed on the Rayleigh–Ritz principle for the *scanned* k-dependent determinant and does not partition the argument between the locked and scanned postings; a targeted consultation returned no statement that the fixed-scale metric-free problem bounds from above. We therefore claim it.

[ESTABLISHED, from Avery] The mechanism is Avery's, not ours. The helium ground state needs *in-out* radial correlation, which variational treatments supply through split-shell $(1s\,1s')$ functions carrying two independent exponents; a Goscinskian $1s^2$ configuration places both electrons at $n = 1$ and therefore at the identical exponent Q_ν/n_j—for *any* weighting potential, since a separable V_0 still assigns one β_ν per configuration. Excited configurations obtain two scales free from $n_a \neq n_b$, and the state-dependence this predicts is measurable: the posing cost is 4.21 mHa for the ground state against 0.98 mHa for $2\,^1S$ at $K = 105$, and the ratio in fact *narrows* slowly with basis size — 5.17 at $K = 36$, 4.66 at 78, 4.29 at 105 — while the absolute separation widens (2.84 → 3.23 mHa) and then flattens. The floor is a ground-state pathology rather than a property of the method.

[INTERNAL THEOREM] *Selection cannot rescue the locked posing.* Because T' is a matrix of pure numbers, its entries do not depend on which *other* configurations are present, so the secular matrix of any subfamily A is *exactly* the corresponding principal submatrix of M (verified bit-identically). Cauchy interlacing then gives $\lambda_{\max}(M_A) \le \lambda_{\max}(M)$, and since $E = -p_\kappa^2/2$

with $p_\kappa = \lambda_{\max} > 0$,

$$E(A) \;\geq\; E(M) \qquad \text{for every sub-family } A \subseteq M, \quad (5)$$

the families being nested in $n_{\max}$ so that E decreases monotonically along the ladder. No choice of configurations, however optimized, beats the family it is drawn from. **[MEASURED]** The best 102 by ground-state weight out of a $K = 244$ pool returns 7.25 mHa against that pool's own 7.06 mHa — *above* it, as Eq. (5) requires — while of 200 random 102-subsets the best reaches only 13.9 mHa and the worst 903 mHa — 58% of them omit the dominant $1s^2$ configuration entirely. Note what supplies the bound: it is the same pure-number property that makes the encoding attractive. Were T' basis-dependent, sub-families would not be principal submatrices and interlacing would not apply.

[OBSERVATION] A comparison to the −2.90250 Ha reached with 102 optimized Coulomb–Sturmian configurations [2] is therefore *withdrawn*, and on stronger grounds than a mismatch of numbers: by Eq. (5) that figure (1.2 mHa short) *cannot* have come from the locked-scale Goscinskian posing, at any K or under any selection — our largest computed family stands at 6.82 mHa and the ladder is monotone. It matches instead our scale-optimized ladder (1.64 mHa at $K = 100$). **[OBSERVATION]** Consultations of the primary canon, relayed, settle the attribution — and one of them supplies an external test of Eq. (10). The posing was identified first: the scaling parameter *scanned as a free variational parameter and minimized*, unlinked from the output eigenvalue, with the ground state's difficulty attributed there, as here, to $1s^2$ forcing both electrons to $Q_\nu = p_\kappa/\sqrt{2}$. The same account also described that calculation as evaluating a *metric-free* matrix at each step — which Eq. (10) forbids, since projecting a basis built at E_{basis} against a target E_{out} leaves the L^2 overlap with coefficient $(E_{\text{basis}} - E_{\text{out}})$, and unlinking the scale is precisely what makes that coefficient nonzero. Put back to the source together with the projection, the derivation was accepted and the description corrected: the calculation was an *ordinary variational* configuration interaction, $HC = E\,SC$, and the metric-free cancellation belongs exclusively to the locked isoenergetic method. We record the attribution as secondary-source — it is not a quotation, and a primary-source check remains the one thing that could overturn this section — but the episode is worth stating for what it is: an external description of a calculation as simultaneously unlinked and metric-free, posed against Eq. (10) and withdrawn. The machinery is correct, including the higher-ℓ angular correlation (Gaunt / Wigner-3j multipole coupling).

The decisive quantity is the block-encoding cost. **[MEASURED]** The entrywise 1-norm of the secular matrix, $\|M\|_1 = \sum_{\nu'\nu} |M_{\nu'\nu}|$ (an upper bound on the achievable LCU λ), grows *more slowly than the matrix dimension* K,

$$\|M\|_1 \sim K^{0.82} \quad (\text{full } s{+}p{+}d{+}f,\ \text{helium},\ K = 74\text{–}164), \quad (6)$$

the opposite of the naive inflation Eq. (1).

Three qualifiers travel with that number, and none is cosmetic.

[MEASURED] *First, it is not a stable exponent.* The fit is over a window, and the window's value is not the object's behaviour. Continuing the same family, the *local* slope falls monotonically: 0.827, 0.818, 0.810, 0.802, 0.794, 0.787, 0.781, 0.776, 0.771, 0.766 across $K = 100$–514, with no plateau at the largest basis we can reach. A two-parameter $c\,K^p \log K$ form fits the data an order of magnitude better than any pure power law, for every norm family we measured. The honest statement is that $\|M\|_1$ grows sublinearly over every computable basis and moves *away* from linear as the basis grows; any single exponent is a property of the window it was fitted in.

[MEASURED] *Second, it is a helium number.* T' is Z-independent by construction, so $\|M\|_1$'s exponent is a mixture whose weight moves with the nuclear charge: the same window gives 0.947 at $Z = 1/2$, 0.819 at $Z = 2$, and 0.717 at $Z = 30$. Every exponent in this paper is quoted for helium.

[MEASURED] *Third — and this is the substantive finding — the sublinearity is a property of how the basis is grown, not of the isoenergetic construction.* The family used here holds $l_{\max}$ fixed and adds higher-n configurations, whose couplings decay as n^{-2}. Grown the other way, as full hydrogenic shells with $l_{\max} = n_{\max} - 1$, the same secular construction gives $\|T'\|_1 \sim K^{1.07}$ — superlinear — while the *total* exponent rises but stays below 1 (local slopes $0.867 \to 0.911$ across the rungs from $K = 35$ to $K = 220$; the first slope lying wholly inside $K = 56$–220 is 0.879, and a global fit over that window is 0.893). A superlinear block and a rising total are therefore both real for a different growth rule; but no measured full-shell point makes the total itself superlinear, and we do not claim one. What Eq. (6) measures is the fixed-$l_{\max}$ rule, and the cost advantage belongs to that rule rather than to the method.

[MEASURED] *Which block carries it.* Splitting $M = \mathrm{diag}(ZR_\nu) + T'$ and fitting each over $K = 74$–340 on a converged radial domain:

$$\|T^0\|_1 = Z\sum_\nu R_\nu, \qquad \|T'\|_1 \sim K^{0.88}, \qquad \|T'\|_1^{\text{off}} \sim K^{0.94}. \quad (7)$$

The three are related by the exact identity $\|M\|_1 = \|T^0\|_1 - \|\mathrm{diag}\,T'\|_1 + \|T'\|_1^{\text{off}}$. **Neither T' grouping is superlinear**: the full block's local slope crosses 1 near $K \approx 37$ and falls to 0.805 by $K = 514$; the off-diagonal part crosses near $K \approx 91$ and falls to 0.841. The nuclear diagonal is nonetheless the slowest-growing block, and its share of the norm falls monotonically (64.9% at $K = 74$ to 57.0% at $K = 340$), so Eq. (6)'s exponent sits between the two and drifts with their mixture. This is the same structural fact Sec. VI reaches from the other side: the advantage rides on the clean single-center diagonal T^0, which is exactly what two centers destroy. Angular structure dilutes but does not break the contrast — the

s-only exponent 0.73 rises to 0.82 with p, d, f present, on matched windows.

[SYMBOLIC] *The nuclear diagonal is not a power law at all.* $\|T^0\|_1 = Z\sum_\nu R_\nu$ requires no two-electron integral and no radial grid: it is a closed-form sum over the configuration list, with $K = 2N^2 - 4N + 4$ exactly for $l_{\max} = 3$. Its asymptotics are

$$\|T^0\|_1 \;\sim\; Z\sqrt{2K}\,\ln(K/2), \tag{8}$$

so its log–log slope is $\frac{1}{2} + O(1/\log K)$ — asymptotic exponent 1/2, not a fitted power. Evaluated on the exact sum the slope runs 0.709 at $K = 20$, 0.705 at 164, 0.662 at 12484 and 0.603 at 498004, still falling. A fit inside $K = 74$–164 returns 0.705; that is what the window yields, not what the object does. Eq. (8) is the one leg of this section's *scaling* measurements that is derived rather than measured, and it is grid-independent by construction.

[MEASURED] *What the sublinearity costs.* The fixed-$l_{\max}$ family does not converge to the exact energy. On the converged ladder its ground-state error is 8.04 mHa at $K = 74$ and 6.78 mHa at $K = 514$; fitting $\Delta E(K) = \Delta E_\infty + bK^{-q}$ gives $\Delta E_\infty = 6.44$ mHa (residual < 0.002 mHa), with 6.446 mHa still predicted at $K = 10^5$. Seven times the basis closes 16% of the deficit. That single full-range fit is dominated by its small-K end, however, and because this fit family approaches the floor *from below* it is a lower estimate rather than a central one; resolving it window by window and pairing it with a model-free extrapolation brackets the floor at [6.47, 6.62] mHa (below), and it is the bracket we quote. Either way the floor is $\approx 4\times$ chemical accuracy (1.594 mHa).

[INTERNAL THEOREM] *Where the floor comes from.* It is neither angular truncation nor a deficiency of the span. Write the configurations at a free global scale λ, which the Goscinskian construction fixes to p_κ. Because each Φ_ν is hydrogenic at $Q_\nu = \lambda/R_\nu$, the one-body Coulomb metric is exactly diagonal; written at unit scale ($\lambda = 1$, the convention used for T below, so that at general λ the diagonal reads λR_ν),

$$W_{\mu\nu} \equiv \left\langle \Phi_\mu \middle| \sum_j r_j^{-1} \middle| \Phi_\nu \right\rangle = R_\nu\,\delta_{\mu\nu}. \tag{9}$$

[SYMBOLIC] This is exact, not numerical. The diagonal is $W_{\nu\nu} = Q_\nu \sum_j n_j^{-2} = Q_\nu R_\nu^2 = R_\nu$ from $\langle 1/r\rangle_n = Q/n^2$. Off the diagonal there are three cases. If $l_\mu \neq l_\nu$ the integral vanishes by angular orthogonality. If the l agree and $R_\mu \neq R_\nu$, apply $(T - E)$ to the two Sturmian equations and use its hermiticity: the cross terms give $(Q_\mu - Q_\nu)\,W_{\mu\nu} = 0$, so $W_{\mu\nu} = 0$ whenever the weighted charges differ. The remaining case, $R_\mu = R_\nu$ at equal l, does not occur below $n_{\max} = 35$ (the largest basis computed here is $n_{\max} = 17$), and when it does occur W still vanishes: equal R forces equal charge, and two distinct configurations sharing a charge have disjoint n-multisets, so the radial factors are orthogonal. The identity is therefore unconditional on this family, and the entrywise agreement to 5×10^{-11} is a check on the implementation rather than the evidence for the claim. *This is the same potential-weighted orthogonality that Sec. VI invokes for the molecular problem — automatic for eigenvectors of a common Sturmian problem with distinct β — read back in its atomic special case.* Hence the kinetic matrix at unit scale is $T = \mathbb{1} - S/2$, and the variational problem over the same span is $H(\lambda)C = E\,SC$ with $H(\lambda) = \lambda^2(\mathbb{1} - S/2) + \lambda(-Z\,\mathrm{diag}\,R_\nu + G)$ and $G_{\mu\nu} = \langle\Phi_\mu|r_{12}^{-1}|\Phi_\nu\rangle$. Putting $E = -\lambda^2/2$ cancels every S term *identically* and leaves $(Z\,\mathrm{diag}\,R_\nu - G)C = \lambda C$, which is Eq. (3). The metric-free posing is therefore the variational problem of the same span, evaluated at the one scale where the L^2 metric drops out:

$$\text{metric-free} \iff E = -\lambda^2/2 \iff \lambda = p_\kappa = \sqrt{-2E}. \tag{10}$$

[MEASURED] That lock is not the variational optimum. Freeing λ over the identical span reaches 0.15 mHa of the exact s-limit at $K = 136$ (s-only, where -2.879029 Ha is known independently) against 4.40 mHa locked, and 1.28 mHa of the exact energy at $K = 130$ (*spdf*) against 7.46 mHa locked. The span is not the limitation; the lock is. Two corollaries. The variational bound is automatic rather than fortunate, and the argument gives more than the bound. Because W is diagonal, $H(\lambda) + \frac{1}{2}\lambda^2 S = \lambda\,(\lambda\mathbb{1} - M)$ *exactly*; S is positive definite, so by Sylvester's law of inertia the number of roots of the pencil lying strictly below $-\frac{1}{2}\lambda^2$ equals $\#\{k : \lambda_k(M) > \lambda\}$. Taking $\lambda = \lambda_{\max}(M)$ that count is zero, which is the bound; taking $\lambda = \lambda_k$ it is exactly k, which is the root-by-root correspondence the excited-state results of this section rely on. Neither step needs $E_{\rm iso}$ to be identified as the lowest root in advance. And freeing the scale forfeits the encoding advantage entirely. On the s-only comparison ladder ($K = 21$–136; note this is a different sector and window from the headline exponent of Eq. (6), so the four numbers below are quoted as one matched set) the locked 1-norm's $K^{0.72}$ becomes $K^{1.95}$ for $H(\lambda^*)$ and $K^{2.75}$ for the whitened $S^{-1/2}HS^{-1/2}$, with $\mathrm{cond}(S) \sim K^{0.94}$ — a factor 5×10^3 at $K = 136$. *Metric-free posing and accuracy floor are one fact, not two — both are the scale lock, and the* 1-*norm advantage* over the free-scale alternative *is its third consequence. The sublinearity itself is not: that belongs to the growth rule.*

[ESTABLISHED, from Avery] *Why the ground state in particular.* Helium's ground state needs *in-out* radial correlation, supplied variationally by split-shell $1s\,1s'$ functions carrying two independent exponents. A Goscinskian $1s^2$ configuration places both electrons at $n = 1$ and hence at one exponent Q_ν/n_j — for *any* weighting potential, since a separable V_0 still assigns a single β_ν per configuration. Excited configurations obtain two scales free from $n_a \neq n_b$. **[MEASURED]** The state-dependence this predicts is large, and it is free: $\|M\|_1$ is a property of M, not of which root is extracted, so every ^1S state costs the same to block-encode

and only the delivered accuracy differs. At the largest basis where both roots were computed, $K = 452$, the ground-state error is 6.82 mHa (4.28× chemical accuracy) against 1.716 mHa (1.08×) for $2\,^1S$. Reference-free — the posing cost $E_{\rm iso} - \min_\lambda E_{\rm var}$ requires no known limit — the ^1S ladder — s-only at this rung, where the $K = 452$ pair above is full $s+p+d+f$ — at $K = 105$ reads 4.21, 0.98, 0.3235 and 0.1251 mHa across the first four roots — reductions of 4.3×, 3.0× and 2.6× per rung, themselves shrinking. (The last two are printed to four figures because at two the quoted ratio no longer reproduces.) **[OBSERVATION]** The shrinking Rydberg gaps do not spoil this at chemical accuracy: resolving the k-th root needs $\epsilon_p = \epsilon_E/p_\kappa(k)$, which runs 6.6–$7.9\times10^{-4}$ across the first four roots, against p_κ gaps of 0.336, 0.041, 0.014 and 0.006; requiring half a gap of separation gives margins of 254, 27, 8.6 and 3.9 — so the query count $\|M\|_1/\epsilon_p \approx 2\times10^5$ is state-independent through $4\,^1S$. **[MEASURED]** The remaining cost of an interior root is state preparation, and it is modest. In the L^2 metric the overlap of the dominant single configuration with the true root runs 0.992, 0.798, 0.864 and 0.889 across the first four roots, so $2\,^1S$ — not the deeper ones — is the hardest, at 1.25× the ground state in rotations and 1.54× in repetitions; two configurations already reach 0.99. Interior roots become *easier* again, so the driver is mixing at the bottom of the Rydberg series rather than spectral depth. One practical caveat: the dominant configuration does not track the spectroscopic label — the root at He $3\,^1S$ is dominated by $(l, n_a, n_b) = (0, 1, 4)$ — so a preparation heuristic keyed to the physical principal quantum number selects the wrong configuration.

[OBSERVATION] *How well the floors themselves are known.* They are bracketed rather than pinned, and the dominant uncertainty is not the one we first looked for. Across 21 fit windows the free-floor value drifts by only 1.3% (ground) and 0.4% ($2\,^1S$), and drifts *upward*: the fit approaches from below. A model-free Shanks extrapolation descends from above, so the two together bracket the floor at $[6.47, 6.62]$ and $[1.647, 1.676]$ mHa — the lower endpoint from the fit, the upper from Shanks. Window drift is therefore not the exposure; *model family* is. A two-parameter $c + b/\ln K$ form would place the $2\,^1S$ floor *below* chemical accuracy. It is rejected on two grounds rather than one: it fits 96–342× worse in RMS, and — unlike the three-parameter form, whose limit is stable to three decimals across every window (1.032, 1.033, 1.033× chemical accuracy) — *its own limit is window-unstable*, running 0.69 → 0.79 → 0.82× as the window tightens. A model whose asymptote moves with the window is not asserting an asymptote. In the state-dependence comparisons below we therefore quote measured ladder endpoints rather than fitted values, and where a fitted floor is quoted we give its bracket rather than a single value. We record that "$2\,^1S$ saturates above chemical accuracy" remains a model-selection conclusion rather than a measurement.

[OPEN] Two questions replace the one this section used to raise. Whether a weighting potential of different *radial shape* can reposition the exponent distribution enough to lower the ground-state floor while preserving Eq. (9) — a constant rescaling cannot, since $\beta_\nu Z_w = p_\kappa/R_\nu$ eliminates Z_w identically, so the lever must change the shape of V_0 and not merely its strength. That lever has a named price, and it is not small: a screened or mean-field V_0 improves the physical quality of each configuration, but losing the $1/r$ form breaks the mapping to hyperspherical harmonics under the Fock projection, and with it the closed-form evaluation of the multi-center and inter-electron integrals — the property this entire encoding rests on. The open question is therefore sharper than "does a better V_0 exist": it is whether any V_0 of different radial shape *preserves the Fock mapping*. If none does, the trade is a wall with a mechanism rather than unexplored ground. And whether the excited-state operating point survives the state-preparation cost that Eq. (10) says nothing about. As it stands the family is best read as a *fixed-accuracy tier whose accuracy depends on the target state*: sublinear cost at 6.82 mHa for the ground state and 1.72 mHa for $2\,^1S$ — both measured at $K = 452$ — bought at the same price.

Scope and comparison. The "sublinear" here is a property of the LCU 1-norm $\|M\|_1$ of a *fixed*-size atomic secular matrix as the configuration count K grows; it is a distinct quantity from the sublinear *gate complexity in the number of plane-wave basis functions* of Babbush *et al.* [23]. Its exponent is in K at *fixed* atomic size — a different axis from the system-size scaling of the electronic-structure 1-norm literature, where the best documented basis-choice lever, localized orbitals [22], reaches $\sim N^{1.34}$ in the orbital count and is still superlinear; the two are a regime contrast, not a like-for-like exponent comparison. Two further caveats: the construction uses L^2-normalized configurations rather than the exact potential-weighted normalization, and K is the matrix dimension, not the qubit count. Given that the exponent has already moved once under a change of numerical domain, we no longer assert that the normalization would leave it fixed; that is an open check.

Provenance of the numbers in this section. **[MEASURED]** All values above are computed on a radial domain satisfying $R_{\max} \geq 3n_{\max}^2$, the convergence rule established in Appendix A, and were obtained by two independent routes — exact grid-free evaluation of the Slater integrals in closed form, and converged quadrature on a graded mesh — agreeing to four decimal places on every leg. An earlier version of this section reported $K^{0.84}$, $\|T'\|_1^{\rm off} \sim K^{1.05}$ and an exponent "trending toward 1" [retracted 2026-09-08: p60-tprime-superlinear]. Those were computed on a fixed 60-bohr radial domain, which truncates configurations whose orbitals extend to 100–200 bohr; because the truncation error grows with K (by a factor 1.011 → 1.205 across the fitted range) it inflated the fitted slope and reversed its apparent drift. The corrected values are reported here.

V. THE QUANTUM ALGORITHM AND ITS NOVELTY

Equations (3)–(6) describe an unusually clean target for quantum eigenvalue estimation: block-encode the fixed, bounded, pure-number matrix M; estimate its eigenvalues p_κ by qubitization or phase estimation [9]; return the energies $E = -\frac{1}{2}p_\kappa^2$. **[OBSERVATION]** A targeted prior-art search finds no existing quantum algorithm combining (i) a Sturmian/hyperspherical basis, (ii) a quantum eigenvalue routine, and (iii) the isoenergetic inversion (fix E, solve for a scale). Coulomb–Sturmian bases are classical in the literature [5]; quantum electronic structure uses Gaussian and plane-wave bases; and quantum generalized-eigenvalue solvers [10, 11] run the forward direction. The closest relative [11] scans a candidate eigenvalue as a singularity test, which is not the isoenergetic advantage. The atomic algorithm therefore appears novel.

The two documented cost risks for such a scheme are, moreover, *absent in the atomic case*. The metric-conditioning multiplier of block-encoded generalized eigenproblems [10, 12] does not apply, because Eq. (3) carries no metric. The outer energy search that would otherwise multiply the cost does not arise, because the eigenvalues are the energies. What remains is a single figure of merit, the 1-norm of M, which Eq. (6) measures to grow more slowly than the matrix dimension over every computed basis (with the caveats on window and mechanism recorded there). We stress that this is a resource analysis, not a hardware demonstration.

VI. THE MOLECULAR FRONTIER: SHIBUYA–WULFMAN CONDITIONING

[INTERNAL THEOREM] *The general-V_0 form, and what the metric is.* Before specializing, the general statement costs two lines and is worth having, because it says what the molecular metric actually is. Projecting $\langle\Phi_\mu|H - E|\Phi_\nu\rangle = 0$ and substituting the Sturmian equation on the ket, $(-\frac{1}{2}\sum_j \nabla_j^2 - E)|\Phi_\nu\rangle = -\beta_\nu V_0|\Phi_\nu\rangle$, gives

$$\sum_\nu \Big(\langle\Phi_\mu|V|\Phi_\nu\rangle - \beta_\nu\langle\Phi_\mu|V_0|\Phi_\nu\rangle\Big)C_\nu = 0, \qquad \text{that is} \qquad \mathbf{V}\,\mathbf{C} = \mathbf{V}_0\mathbf{B}\,\mathbf{C}, \tag{11}$$

$\mathbf{B} = \mathrm{diag}(\beta_\nu)$. Every L^2 overlap cancels *identically* — for any local V_0, and whether or not the configurations are orthonormal. **[MEASURED]** Atomically $\langle\Phi_\mu|V_0|\Phi_\nu\rangle = -ZR_\nu\delta_{\mu\nu}$ and $\beta_\nu Z = p_\kappa/R_\nu$, so $\mathbf{V}_0\mathbf{B} = -p_\kappa\mathbb{1}$ and Eq. (11) collapses to Eq. (3): verified to 6×10^{-10}, with the reconstruction $Z\,\mathrm{diag}\,R_\nu - G$ bit-identical to the assembled M. **[OBSERVATION]** Molecularly $\mathbf{V}_0$ is *not* diagonal, so the matrix on the right-hand side is the weighting potential itself — not the identity, and not the L^2 overlap. That identifies the Shibuya–Wulfman matrix rather than merely naming it: for $V_0 = -\sum_A Z_A/|\mathbf{r}-\mathbf{R}_A|$ its off-diagonal entries are exactly the cross-center nuclear-attraction integrals, which is what `geovac/shibuya_wulfman.py` computes. The molecular metric is not an extra structure the method acquires under generalization; it *is* V_0. *Provenance:* the Shibuya–Wulfman integrals and their momentum-space evaluation are Avery's; the identification of the resulting matrix *as* the V_0-weighted overlap is our formalization — a targeted consultation could not locate that sentence in the texts, only the integrals it would describe.

[ESTABLISHED, from Avery] Molecularly there is no closed-form counterpart to $\beta_\nu Z = p_\kappa/R_\nu$. Because V_0 is multi-center, the weighting factors are obtained as the exact eigenvalues of the one-electron molecular Sturmian problem $[-\frac{1}{2}\nabla^2 - E + \beta v_0]\phi = 0$ at fixed E, with a many-electron configuration's β_ν following from the isoenergetic condition on that spectrum. This is what makes the molecular orbitals of Sec. VI S_{SW}-orthonormal *by construction* rather than by arrangement: potential-weighted orthonormality is automatic for eigenvectors of a common Sturmian problem with distinct β, and S_{SW} is the potential doing the weighting. Two facts the paper had been carrying separately are one fact.

The clean atomic result does not fully transfer to molecules. The multi-center one-electron isoenergetic problem is again *generalized*,

$$\sum_\mu \big[\, W_{\mu'\mu} - k\, S_{\mu'\mu}\,\big] C_\mu = 0, \tag{12}$$

with S the Shibuya–Wulfman matrix [3], whose condition number multiplies the block-encoding cost: inverting or square-rooting S by quantum singular-value transformation costs a polynomial degree growing with κ (Sec. VI A) [13, 14]. In Hermitian form $S_{\mu'\mu} = (2k^2)^{-1}\langle\nabla\varphi_{\mu'}|\nabla\varphi_\mu\rangle + \frac{1}{2}\langle\varphi_{\mu'}|\varphi_\mu\rangle$.

[MEASURED] On a two-center s-orbital Coulomb–Sturmian basis we find: (i) the intra-center block of S is *exactly* the identity—in the momentum-space evaluation, diagonal $\{1,1,1\}$ with off-diagonal $\sim10^{-16}$ (the 2D-grid values $\{1, 0.999, 0.998\}$ are discretization error)—where the L^2 overlap's intra-center block is merely non-orthogonal (cond 5.8, growing with basis), so the within-center disaster of Sec. II is genuinely absent from S; (ii) across two centers S is uniformly better-conditioned than the L^2 overlap and markedly better at larger separation:

R (bohr)	basis	cond(L^2)	cond(S)
1.4	6	39.1	30.1
2.0	6	20.9	15.4
4.0	6	11.1	**4.4**

but (iii) cond(S) still grows with basis size—*polynomially*, cond(S) $\sim N^{1.85}$, essentially the same rate as the L^2 overlap's $N^{1.71}$ (decisively power-law, not exponential), with the largest eigenvalue bounded, $\lambda_{\max} < 2$, and all the growth in $\lambda_{\min} \to 0$ as the basis approaches linear dependence—driven by the inter-center coupling

(off-block magnitude $\sim$ 0.4–0.6), so the improvement over L^2 is only modest at bonding distance. All three cross-check across a 2D position-space grid, the closed-form momentum-space quadrature, and a direct three-dimensional momentum integral (agreement to $\sim 10^{-10}$).

[SYMBOLIC + MEASURED] The conditioning law has an exact derivation rather than being a fit (the derivation is prior art — see the **[PRIOR ART]** paragraph below — and what we claim is the identification), and the fitted exponent is a window reading. Because the intra-center block is exactly the identity, $S = \left[\begin{smallmatrix} I & C \\ C^\intercal & I \end{smallmatrix}\right]$ has spectrum $\{1 \pm \sigma_k\}$ with σ_k the singular values of the cross block (the cosines of the principal angles [17, 18] between the two center subspaces), so $\mathrm{cond}(S) = (1+\sigma_{\max})/(1-\sigma_{\max})$ *exactly*. This chain is classical and we use rather than derive it: the principal angles are Jordan's [24] (their computation is [25]), the spectrum $\{1 \pm \sigma_k\}$ is the Jordan–Wielandt form of the off-diagonal block, and the resulting condition number is the two-block CBS constant of subspace-correction theory [26]. In the sine basis the cross block is the finite section of a multiplication operator with symbol $j_0(kR\cot(\chi/2))$, whose supremum 1 is attained at $\chi = \pi$; the band-limited concentration rate at a quadratic symbol maximum gives

$$1 - \sigma_{\max} \;=\; \frac{(kR)^2}{24}\,\frac{\pi^2}{n^2} + o(n^{-2}), \tag{13}$$

(n orbitals per center), i.e. the conditioning exponent is exactly 2 asymptotically, with the single-parameter collapse $(1-\sigma_{\max})\,(n/kR)^2 \to \pi^2/24$ (confirmed to $\sim$1% at $n = 160$; the leading-order derivation is asymptotic, the remainder is measured, not bounded). The fitted $N^{1.85}$ above—and the $N^{1.81}$ of the same quantity at $R = 2$ in the sprint record—are pre-asymptotic slopes of this one law read on different windows.

[PRIOR ART] Eq. (13) is not new *as an asymptotic law*, and we claim only the identification (whose smoothness hypothesis our symbol does not meet — see below; that is a caveat on the proof we can cite, not a claim on our side of the line). It is the Kac–Murdock–Szegő extreme-eigenvalue asymptotic [46]: writing the relevant symbol in the normal form $|1-t|^{2\alpha} b(t)$ used by Böttcher and Widom [47], one has $\lambda_{\min} \sim (c_\alpha/n^{2\alpha})\, b(1)$, with $c_1 = \pi^2$ due to Kac, Murdock and Szegő. Our symbol is the case $\alpha = 1$: with $\theta = \pi - \chi$, $1 - j_0(kR\cot(\chi/2)) = (kR)^2\theta^2/24 + O(\theta^4)$, so $b(1) = (kR)^2/24$ and $c_1 b(1)/n^2$ reproduces Eq. (13) constant and all; the conditioning exponent 2 is the order of the symbol's maximum and nothing else. What is ours is the *identification* — that the two-center Shibuya–Wulfman metric in the sine basis is such a finite section, Toeplitz minus Hankel since $\langle n|a|m\rangle = c_{n-m} - c_{n+m}$, with symbol $j_0(kR\cot(\chi/2))$. It is *not* the reading of the Shibuya–Wulfman operator as a translation: that is prior art on three counts, and earlier versions of this paper claimed it. Shibuya and Wulfman already build the molecular p_0 operator from the united-atom one as a sum over the nuclei, one unitary transformation per center [40], which is the structure in substance. (We paraphrase rather than quote: the bibliographic record is verified at source, but the 1965 abstract is behind a paywall and three independent attempts failed to reach its text, so an earlier verbatim quotation here is withdrawn as unverified rather than asserted.) Wulfman and Takahata gave the explicit continuous-group formulation two years later, in terms of the Lie algebras of E_4, R_5 and $O(4,1)$ [37]; and Red and Weatherford derived the general formula for that matrix in a Coulomb–Sturmian basis [38]. That the angular average $j_0(pR)$ becomes trivial at $p = 0$ is then a property of the symbol, which is the part we are entitled to. The physical reading follows: a basis complete enough to carry wavelengths longer than R cannot tell the two centers apart. Equivalently, $1 - \sigma_{\max} = \frac{1}{6}(R/L_{\max})^2$ with $L_{\max} = 2n/(\pi k)$ the longest wavelength the truncated basis carries, so the degeneracy switches on exactly when $L_{\max}$ exceeds the bond length — which a complete basis must eventually do.

[MEASURED] One reading of this degeneracy should be resisted, and the measurement excluding it is cheap. Because the intra-center block is exactly the identity, the A-set is *orthonormal* in the S metric, so the deficit of a displaced Sturmian against the one-center span is directly computable: $\varepsilon_N^2 = 1 - \sum_{i\le N}\langle\chi_i^A, \chi_1^B\rangle^2$ by Bessel's inequality. It does *not* tend to zero. It plateaus at 0.380, 0.696 and 0.907 for $kR = 1, 2, 4$, flat from $N = 16$ to $N = 256$ (`tests/test_paper60_one_direction.py`). The one-center shared-scale set is therefore measurably far from complete *in the molecular metric* — between 38% and 91% of a displaced basis function lies outside its span — so the degeneracy is *not* the statement that one center's span already contains the other's. What $\sigma_{\max} \to 1$ asserts is weaker and more useful: there exists a *combination* in the B-span that the A-span captures, concentrated where the symbol attains its supremum. The near-dependence is *one direction*, not a diffuse property of the basis — which is why the fixed, geometry-independent rank-$(M-1)$ rotation of Sec. VI A removes it at $M = 2$ and for non-collinear $M = 3$, and why the coupling's participation ratio stays small. It also qualifies the motivation offered below for this basis: the Coulomb–Sturmians' completeness at a single scale is completeness in the *atomic* metric, and the molecular problem is posed in V_0, where the one-center set is not complete.

[SYMBOLIC + MEASURED] One residue is worth stating rather than hiding, and it has a closed form. Our symbol does *not* satisfy the smoothness hypothesis under which [47] proves the constant: at the opposite end $\chi \to 0$ it is a chirp, amplitude $\sim \chi/(2kR)$ and phase $\sim 2kR/\chi$. Its coefficient asymptotics need no stationary-phase argument, because the model integral $\int_0^\infty x\, e^{i(2c/x + jx)}\, dx$ is a modified Bessel function: it is DLMF (10.32.10) [34] at $\nu = 2$, continued from that formula's $|\mathrm{ph}\, z| < \pi/4$ to $|\mathrm{ph}\, z| = \pi/2$, and DLMF (10.40.2), whose stated validity $|\mathrm{ph}\, z| \le \frac{3}{2}\pi - \delta$ covers that sector

directly, gives

$$|c_j| = (2\pi)^{-1/2} 2^{-3/4} (kR)^{-1/4}\, j^{-5/4} \left|\sin\!\left(2\sqrt{2kRj} + \tfrac{\pi}{4}\right)\right| + o(j^{-5/4}), \tag{14}$$

constant and phase included, verified over $j = 64$–65536 at three values of kR by three independent quadrature routes, sign pattern included. Two corrections come with it. The $\pi/4$ is the branch phase of the $(\pi/2z)^{1/2}$ prefactor, *not* a stationary-phase signature. And $\sum_j |c_j|$ *converges* — $5/4 > 1$, so the symbol is in the Wiener algebra; what diverges is $\sum_j j|c_j|$, which is the hypothesis Böttcher–Widom needs and a different condition entirely. The constant nevertheless holds. Relatedly, the $\sim$1% figure quoted above is structure rather than scatter — the relative residue halves under each doubling of n, falling $0.134 \to 0.010$ across $n = 10 \to 160$ at $kR = 2$ — and it is *dominated by the grid convention*: the τ-algebra statement places the eigenvalues at $j\pi/(n+1)$, and collapsing with $(n+1)^2$ in place of n^2 moves the residue at $n = 160$, $kR = 2$ from -0.99% to $+0.26\%$, a fourfold reduction with a change of sign. A genuine $O(1/n)$ term survives in both conventions. Whether a weaker hypothesis covers this symbol class is left open.

[SYMBOLIC + MEASURED] The two transcendentals of this section are tagged here against Paper 18, and the tagging is a statement of *provenance* — where each constant comes from — nothing more. The π^2 of Eq. (13) is the Kac–Murdock–Szegő constant c_1, and its reading as an eigenvalue is already in the source cited above: Böttcher and Widom [47] present c_α as the least eigenvalue of the boundary-value problem $(-1)^\alpha u^{(2\alpha)} = \lambda u$ on $[0,1]$ with $u^{(j)}(0) = u^{(j)}(1) = 0$, which at $\alpha = 1$ is the Dirichlet problem for $-d^2/dx^2$ and gives π^2; their title names the extremal (Wirtinger–Sobolev) inequality that carries it, and the independence of c_α from b is likewise their observation, not ours. The constant is therefore fixed by the order of the symbol's zero together with the finite section, and carries no Bessel content: nothing about j_0 beyond $b(1) = (kR)^2/24$ enters it. We record the same statement in the basis's own variable, where it is directly visible — minimising $\langle\theta^2\rangle$ over $\mathrm{span}\{\sin a\chi\}_{a\le n}$ in the Fock polar angle $\theta = \pi - \chi$ gives $n^2 \min\langle\theta^2\rangle \to \pi^2$ (Richardson 9.86949 against $\pi^2 = 9.86960$), the near-null direction attains that minimum with r.m.s. spread π/n (Richardson 3.14158), and $1-\sigma_{\max} = (kR)^2\langle\theta^2\rangle/24$ holds on it to 0.3% at $n = 320$ across $kR = 0.5$–4. *This is a change of representation and not an independent route: it is the $b \equiv 1$ case of the same theorem, and the minimiser is the Dirichlet ground state in the band index.* It is worth writing that identification out, because it carries the *antipodal parity* and is false without it: the basis index is χ while the Dirichlet mode is natural in θ, and $\sin(a\chi) = (-1)^{a+1}\sin(a\theta)$, so the minimiser is $c_a \propto (-1)^{a+1}\sin\!\big(\pi a/(n+1)\big)$. Measured at $n = 320$, that mode correlates with the band minimiser to $1 - 2\times10^{-7}$ and with the near-null direction of the cross block to $1 - 7\times10^{-7}$, while the unalternated $\sin(\pi a/(n+1))$ is *exactly orthogonal* to both. The same factor is the caution attached to the contraction reading of j_0, where the antipodal Mehler–Heine limit converges along parities, rather than outright. The $(2\pi)^{-1/2}$ and the $\pi/4$ of Eq. (14) have a different provenance — the normalisation and the branch phase of a Bessel asymptotic at the $p \to \infty$ pole. Under Paper 18's taxonomy both constants are calibration-tier, sub-mechanism M2: the $\pi^2\cdot\mathbb{Q}$ half of that ring for the first, an M2-side $\sqrt{\pi}$-type normalisation for the second.

What this does not license. An earlier version of this paragraph read the provenance as *predicting* removability — the first price being "a property of the matrix, which a preconditioner reaches" and the second "a property of the symbol, which no congruence can touch" [retracted 2026-09-12: p60-removability-corollary]. Both halves are wrong. The preconditioner of Sec. VI A is built *from the symbol*: its matching polynomial is chosen precisely to share the symbol's zero [45]. And preconditioning *is* a congruence of the finite section, one that replaces the symbol by f/g. The defensible statement concerns the symbol on both sides and turns on the *kind* of feature rather than on the origin of its constant: a banded congruence multiplies the symbol by a trigonometric polynomial, which can cancel a zero of finite order but cannot alter a decay or smoothness class. The conditioning pole is a zero of order two; the locality pole is the chirp's $j^{-5/4}$ envelope. That asymmetry, and not the provenance of π^2, is why the first yields and the second does not. *Open, and load-bearing for the split adopted here:* whether the two poles are genuinely independent features or two faces of one object is not settled — our own reading has gone both ways — and this paragraph takes the weaker, measured position.

[MEASURED + PRIOR ART] Reading the symbol as a quotient yields a prediction, it survives its own control, and it is *known* — in a stronger form than the one measured here. Ahmad, Al-Aidarous, Alrehaili, Ekström, Furci and Serra-Capizzano [41] give the eigenvalues of exactly this preconditioned pencil, and for matrices in the τ (DST-I) algebra their result is an identity rather than an asymptotic: the eigenvalues are the ratio symbol sampled on the grid, with the weight cancelling identically. *Our Toeplitz-minus-Hankel form does not put our matrices inside it*: τ membership requires the coefficient sequence to terminate, which the matching polynomial's does and the chirp symbol's does not, and the DST-I correspondingly leaves 4.1%/2.0%/1.1% off-diagonal weight on the cross block at $n = 16/64/160$ against 10^{-14} for a genuinely-τ matrix. The identity therefore covers the τ *idealisation* of this problem rather than the problem, and the gap is exactly the residue reported above: the grid-sampled τ prediction reproduces $\pi^2/24$ to 9×10^{-6} while the true $\sigma_{\max}$ departs by 0.26%. So the measurement below confirms a theorem instead of establishing one, and the paper claims it as such. Let the metric be deformed by any smooth positive radial weight W on the Fock sphere — the intra-center block becoming the finite section of W and the cross-center block that

of $W\,j_0$. The generalised symbol is then $Wj_0/W = j_0$, which does not see W at all, so both the exponent and the constant should survive unchanged. They do: at $kR = 2$ and $n = 160$ the collapse $(1-\sigma_{\max})(n/kR)^2$ reads 0.4072, 0.4123, 0.4107 and 0.4164 for $W = 1,\ 1+\frac{4}{5}\cos\chi,\ 2+\sin\chi$ and $e^{-\chi}$ against $\pi^2/24 = 0.4112$, with fitted exponents between -1.98 and -2.01 throughout. The control carries the claim: a weight that *vanishes* at the degeneracy, $W = 1+\cos\chi$, moves the constant to 0.828 while leaving the exponent at -1.97, so the agreement above is a measurement and not an insensitivity. *Scope, because the natural overstatement is close by.* This is not independence of the weighting potential V_0: a position-space-local V_0 acts on momentum space by convolution rather than multiplication and therefore leaves this class entirely. What is shown is narrower and still worth having — the law is carried by the translation phase, not by any momentum-space weight multiplying it. It also predicts what a metric *outside* the class should do, and the one measured case agrees: the L^2 overlap degrades with the same exponent but a constant larger by ≈ 1.4, which is what a non-multiplication metric is expected to look like. Whether any position-space V_0 preserves the constant is open.

The same singular spectrum carries the composition wall of the balanced-basis arc: the center-projection commutator is $\|[P_A, P_B]\| = \max_k \sigma_k\sqrt{1-\sigma_k^2}$ — a one-line consequence of the two-subspaces canonical form [48, 49] (the norm identity itself is recorded by Loring [35]), which resolves the pair into 2×2 blocks at the principal angles θ_k, in which the commutator has norm $\sin\theta_k\cos\theta_k$ — which *saturates* at its algebraic ceiling $\frac{1}{2}$ (some σ_k lands near $1/\sqrt{2}$) as the spectrum densifies—the previously reported 0.50 is the saturation value—so the two walls are functionals of one object, with all of the severity information in $1-\sigma_{\max}$ (`tests/test_paper60_sigma_law.py`).

[SYMBOLIC] A third cost does *not* live in this spectrum, and separating it strengthens rather than weakens the obstruction. The underlying algebra is classical rather than ours — that a congruence preserves a block grading exactly when it commutes with it is the symmetry-preservation property of Löwdin's symmetric orthogonalization [28], applied to exactly this multi-center non-orthogonality problem by Slater and Koster [29] and since analyzed for the conditions under which the symmetry actually survives [30]; in operator terms it is the statement that the block-diagonal matrices form a commutant and are therefore inverse-closed. What is ours is only the application, below. If X is invertible and block diagonal with respect to $\mathcal{H} = \bigoplus_\ell \mathcal{H}_\ell$, and $X^\dagger S X$ is block diagonal, then $S = X^{-\dagger}(X^\dagger S X)X^{-1}$ is block diagonal too, being a product of block-diagonal factors. Contrapositively: if S is not ℓ-block diagonal, *no* block-diagonal congruence — Löwdin, canonical, or Cholesky — orthogonalizes it. The two-center metric couples ℓ while preserving m (axial symmetry), so m-selection survives orthogonalization and within-m ℓ-selection cannot, at every condition number > 1: the loss is already present for arbitrarily weak inter-center coupling and does not relax as $\mathrm{cond}(S) \to 1^+$. It is therefore independent of Eq. (13), and the sparsity-destroying orthogonalization excluded below is excluded *structurally*, not on conditioning grounds. That specialization — m survives, within-m ℓ cannot, at every $\mathrm{cond}(S) > 1$ — is the part this paper claims.

The molecular metric cost is therefore *mitigated, not dissolved*: Eq. (12) retains a polynomial conditioning multiplier that is worst where the chemistry lives, and it cannot simply be transformed away—conditioning is orthogonal-invariant, so the momentum-space form (which cancels the Fock Jacobian to a clean one-dimensional quadrature) yields the *identical* matrix and is an exact, grid-free *evaluation* win, not a conditioning improvement. Two genuine conditioning levers do exist: at larger separation $\mathrm{cond}(S) \to 1$ rapidly while the L^2 overlap cannot follow (its intra-center conditioning grows with basis regardless); and gerade/ungerade symmetry adaptation exactly block-diagonalizes S, isolating a gerade sector whose conditioning is flat (cond ≈ 2, flat in basis and R) and confining all ill-conditioning to the ungerade sector—a free $\sim 2\times$ effective-cost reduction in the homonuclear case. The gerade flatness is itself derived: for equivalent centers the blocks are exactly $I \pm C$, and the near-dependent direction has symbol value $+1$, so it lives entirely in $I - C$ (ungerade) while $\mathrm{cond}(I + C) \to 2/(1 + \min_x j_0(x)) = 2.555041\ldots$, an exact constant of the sinc symbol independent of R and of basis size — an instance of the classical fact that a self-adjoint Toeplitz operator's spectrum is the convex hull of its symbol's essential range [27], so that $\mathrm{cond}(I + T(a)) = (1 + \max a)/(1 + \min a)$ (`tests/test_paper60_sigma_law.py`); the "$\kappa \approx 2$" rows below are this constant read at small n.

[MEASURED] This lever is a property of *equivalent* centers, not of symmetry as such, and a probe on water (H_2O, C_{2v}: O + 2H) makes the distinction sharp. For an s-only shared-scale basis the only symmetry action is the H $\leftrightarrow$ H swap (O sits on the C_2 axis, fixed), so the totally-symmetric A_1 block—which holds the ground state—contains *both* the O functions and the symmetric H combinations, and with them the O $\leftrightarrow$ H coupling between symmetry-*inequivalent* centers that the group cannot remove. Building water's three-center SW metric exactly (every s–s block depends only on inter-center distance) and adapting to C_{2v}, the A_1 ground-state block's conditioning *grows*, $\mathrm{cond}(A_1) \sim N^{1.97}$ ($R^2 = 1.000$), essentially the raw metric ($19.9 \to 698$ over $N = 6 \to 36$)—*not* flat like the H_2^+ gerade sector (cond ≈ 2 throughout). Zeroing only the O $\leftrightarrow$ H block collapses it back to cond ≈ 2, pinning that inequivalent-center coupling as the sole driver. The lever therefore does not transfer to a polyatomic with a symmetry-unique heavy center; it is confined to homonuclear-diatomic (equivalent-center) systems and their close analogs. The water block obeys the same exact law: the O $\leftrightarrow$ (H + H) canonical correlations follow Eq. (13) with symbol curvature $c_{\mathrm{sym}} = R_{\mathrm{OH}}^2/24 - R_{\mathrm{HH}}^2/96$ (exponent again exactly 2;

the fitted 1.97 is the same pre-asymptotic reading), and $\mathrm{cond}(A_1) \approx 1.12 \times (1+\sigma_{\max})/(1-\sigma_{\max})$ with a constant prefactor—the canonical correlation is the whole story. The structural reason the symmetry cannot help is now exact: O and H are not exchanged by any group element, so both $1 \pm \sigma$ eigenvector families remain inside A_1 and no block absorbs the divergent factor. A low-rank escape is also closed *structurally*: the coupling's singular spectrum discretizes the range of the symbol (its effective rank grows linearly with basis size, participation ratio ≈ 7 at $N = 36$), and since $1-\sigma_k \approx c_{\mathrm{sym}}(k\pi)^2/n^2$, an exact rank-$r$ (Woodbury) treatment of the top correlations divides the residual κ by $(r+1)^2$ only—a useful constant factor (≈ 40–$75\times$ in d_{inv} terms at $N = 12$–36) but never a change of exponent; flattening would need $r \propto n$, i.e. classically diagonalizing the whole coupling, the sparsity-destroying orthogonalization already excluded.

A. Resource estimate: the gerade lever is the equivalent-center payoff

We convert the measured conditioning into a block-encoding resource count for the one-electron H_2^+ secular equation Eq. (12), at chemical accuracy ($\epsilon_E = 1.6$ mHa) and $R = 2$ bohr. We whiten the generalized problem to a *standard* eigenproblem $\tilde{M} = S^{-1/2} W S^{-1/2}$ whose eigenvalues are the scales k; the metric enters only through $S^{-1/2}$, applied by quantum singular-value transformation with a polynomial approximating $x^{-1/2}$ on $[\kappa^{-1}, 1]$ of degree $d_{\mathrm{inv}} \sim \kappa \ln(\kappa/\epsilon)$ [13, 14], $\kappa = \mathrm{cond}(S)$. This d_{inv} *is* the metric penalty—the quantitative content of the statement that the conditioning multiplies the block-encoding cost [13, 14]. Eigenvalue estimation to precision $\epsilon_k = \epsilon_E/k$ costs $Q \sim (\pi/2)\lambda_{\mathrm{eff}}/\epsilon_k$ qubitization queries [9], with subnormalization $\lambda_{\mathrm{eff}} = k_{\max}$; each query calls one block-encoding of W and d_{inv} of S, so the metric multiplies the query count by $(1 + d_{\mathrm{inv}})$.

[**MEASURED**] The algorithm binds: solving Eq. (12) for H_2^+ by isoenergetic self-consistency (build the basis at scale k, solve $[W - k'S]C = 0$, locate $k' = k$) gives the σ_g ground root $k = 1.475$, i.e. $E_{\mathrm{elec}} = -1.088$ Ha against the reference -1.103 Ha—a 1.3% s-only underbinding (no p-polarization, the expected basis limitation, above the exact value)—and pins $\lambda_{\mathrm{eff}} = k_{\max} \approx 1.7$.

[**RESOURCE MODEL, on the measured κ**] Table I gives the counts. Three readings: (i) the metric-free atomic case ($\kappa = 1$, $d_{\mathrm{inv}} = 0$) is recovered as the zero-penalty limit; (ii) the full SW metric costs $d_{\mathrm{inv}} \sim 10^3$ by $N = 16$; but (iii) the H_2^+ ground state σ_g is *gerade*, and the gerade sector's conditioning is flat ($\kappa_g \approx 2$ across $N = 4$–20), so the ground-state $S^{-1/2}$ costs $d_{\mathrm{inv}} \approx 16$–$18$ *independent of basis size*—a $\sim 60\times$ reduction versus the full metric at $N = 16$, on a register halved to $\lceil \log_2(N/2) \rceil$ qubits. The gerade lever is thus not merely the $2\times$ symmetry factor of the previous paragraph: it delivers the chemically relevant state at a metric penalty that *does not grow with basis size*. This is the equivalent-center payoff (the ground-state lever for homonuclear diatomics and their close analogs, not for a general polyatomic).

TABLE I. Block-encoding resource for the one-electron H_2^+ isoenergetic secular equation Eq. (12) at chemical accuracy ($\epsilon_E = 1.6$ mHa, $R = 2$ bohr), $N = 16$ ($n_{\max} = 8$ per center). d_{inv} is the QSVT degree for $S^{-1/2}$ (the metric penalty); qubits $= \lceil \log_2 N_{\mathrm{dim}} \rceil + n_{\mathrm{phase}} + n_{\mathrm{anc}}$ with $n_{\mathrm{phase}} = 10$. The qubitization query count $Q \approx 2.5 \times 10^3$ is common to every row (it depends on λ_{eff} and ϵ_k, not the metric). A resource model on the measured κ; the $O(1)$ constant in d_{inv} cancels in the cross-metric ratios but not in the absolute counts.

metric	κ	d_{inv}	qubits
atomic (no metric, Sec. III)	1	0	23
SW gerade (σ_g ground)	2.3	18	26
SW full	91.8	1050	31
Gaussian LCAO (ratio 2)	1.99×10^4	3.3×10^5	31

[**RESOURCE MODEL**] Against the same H_2^+ problem in a standard even-tempered s-Gaussian LCAO metric (ratio 2), the SW inverse-degree is 10^2–3×10^2 times smaller (full metric) and 10^3–10^4 smaller (gerade). This is a metric-versus-metric comparison and carries an honest caveat: the Gaussian conditioning is ratio-dependent—a denser ratio-1.6 set (approaching completeness) blows up as cond $\sim N^6$ ($\kappa \sim 10^5$ at $N = 16$), a sparser ratio-3 set grows only as $N^{1.4}$ but leaves coverage gaps and still carries $\sim 10\times$ the SW prefactor. This is the classic Gaussian coverage-versus-linear-dependence trade-off; the Coulomb–Sturmians soften it rather than escaping it, because they are a set complete at one scale *in the atomic metric* — a property the molecular problem does not inherit: the molecular metric is itself polynomially conditioned ($\mathrm{cond}(S) \sim N^{1.85}$ on this window; derived asymptote N^2, Eq. (13)) with a small prefactor. Production Gaussian simulation of course absorbs the overlap classically in a prior mean-field step; the metric penalty priced here is precisely the cost of the on-device angular generation that motivated the Sturmian encoding in the first place.

[**MEASURED**] A third conditioning lever exists; it is stronger than either of the two above *in coverage and on the conditioning axis*, and —unlike the gerade lever—it does not require equivalent centers. It is not uniformly cheaper end to end: where the gerade lever applies its subnormalization is $O(1)$ (the block's least eigenvalue is bounded below by the same $2.555041\ldots$ constant) against this lever's $O(n)$, so on H_2^+ the gerade route remains the cheaper one. Because the ill-conditioning is a symbol *zero* of known order and location, it is removable by preconditioning in the sense of Serra [45]: a trigonometric polynomial sharing the zero bounds the preconditioned spectrum in a fixed interval for every n. The matching polynomial here is $g(\chi) = 2 + 2\cos\chi$, whose quadratic zero sits at $\chi = \pi$ exactly where $1 - \sigma$'s does; in this basis its Hankel part vanishes identically ($n + m \geq 2$ while $g_j = 0$ for $j \geq 2$), so the preconditioner is *exactly* the tridiagonal $\mathrm{tri}(1, 2, 1)$, which the dis-

crete sine transform diagonalizes in closed form (verified against the closed-form eigenpairs to 9×10^{-15}). Writing $G = P^{-1/2}(I - C)P^{-1/2}$, the ungerade conditioning collapses and *stops growing*:

n	cond$(I + C)$	cond$(I - C)$	cond(G)
10	2.383	81.9	2.096
40	2.537	1225.7	2.219
160	2.554	19126.5	**2.229**

The whitening is unaffected in substance: any X with $X^\dagger S X = I$ leaves the generalized spectrum invariant, and $X = P^{-1/2}G^{-1/2}$ is such an X, so the QSVT degree is set by cond$(G) \to 2.23$ rather than by cond(S). On the resource model of Table I that is $d_{\text{inv}} \approx 16$, *flat in basis size*, against 3×10^5 at $n = 160$ untreated. This escapes the $\Theta(\kappa)$ degree floor for QSVT matrix inversion — optimal in $\delta = \kappa^{-1}$ by Theorem 73 of [13], whose Corollary 67 covers x^{-c} for every $c > 0$ and so includes $c = \frac{1}{2}$ — rather than contradicting it: that floor lower-bounds the number of block-encoding queries for an eigenvalue transformation on the *promised spectral interval*, and preconditioning changes that interval — from $[\kappa^{-1}, 1]$ to $[1/2.23, 1]$ — rather than the polynomial approximated on it. The statement that the metric cost is "mitigated, not dissolved" is therefore *too pessimistic on the conditioning axis*, and should be read as applying to the two axes below.

[MEASURED] What preconditioning does *not* buy is locality, and the reason is the second pole. The symbol is pathological at both ends and only one end is a zero: $P^{-1/2}$ cures $\chi = \pi$, while at $\chi \to 0$ the chirp survives untouched. Measured as the bandwidth needed for a fixed relative Frobenius accuracy, $S^{-1/2}$ requires a *fixed fraction* of the matrix ($b/n = 0.72$ at 10^{-2}, flat over $n = 32$–256) whereas $G^{-1/2}$ requires $b = 11 \to 17$ over the same range; but at 10^{-3} the advantage erodes ($b = 27 \to 141$), exactly as algebraic decay predicts. The two are distinguished cleanly by the profile exponent, which for $G^{-1/2}$ sits at -1.19—the chirp's own $-5/4$, *independent of* n—while for $S^{-1/2}$ it drifts with n ($-0.90 \to -0.78$) as the $\chi = \pi$ singularity sharpens. So preconditioning removes one of the two pathologies exactly and leaves the other exactly where it was. The localization class is *not* what blocks a local inverse: with $|c_j| \sim j^{-5/4}$ the metric has polynomial off-diagonal decay of order > 1 in one dimension, which is Jaffard's class [31], and inversion there preserves the decay *provided the operator is boundedly invertible*; inverse-closedness [32] then makes the holomorphic and Riesz calculi agree, so $z^{-1/2}$ is available exactly when $0 \notin$ spec. The escape is the spectrum touching zero, not the decay. Together with the block-structure result above, the honest summary is that the metric's *conditioning* cost is removable, its *locality* cost is capped by the chirp, and its ℓ*-block* cost is not a conditioning effect at all.

[MEASURED] The lever reaches the polyatomic case the gerade lever cannot, and the structural reason is that the degeneracy's *direction* is geometry-independent, and the degeneracy itself is a known one: a kernel matrix whose entries all tend to a common value degenerates to a rank-one all-ones matrix with an $(M - 1)$-fold null space, the *flat limit* of the radial-basis-function literature [33, 43]. What is ours is that the block symbols realize it at a *symbol point* — $\chi = \pi$, i.e. $p = 0$ — rather than in a shape-parameter limit, which is exactly why a fixed rank-$(M - 1)$ rotation removes it. At $\chi = \pi$ every block symbol tends to $j_0(0) = 1$ whatever the separation, so for M centers the matrix symbol degenerates to the rank-one all-ones matrix and its null space has dimension $M - 1$, spanned by a fixed vector set that does not move with geometry or with basis size.

[MEASURED] *The null space is geometry-independent; the rates at which its directions open are not, and that scopes the lever.* Expanding $j_0(pd) = 1 - (pd)^2/6 + O(p^4)$, the order-$p^2$ behaviour on $\mathbf{1}^\perp$ is carried by PD_2P with $(D_2)_{ij} = d_{ij}^2$ and P the projector off the constants. For *collinear* centers $d_{ij}^2 = (i - j)^2 h^2 = h^2\big(i^2 \mathbf{1}^\top + \mathbf{1}(j^2)^\top - 2xx^\top\big)$, and P annihilates the two outer terms from both sides, leaving $PD_2P = -2h^2 Pxx^\top P$ — *rank one.* Exactly one of the $M - 1$ directions then opens at order 2 and the rest at $4, 6, \ldots, 2(M - 1)$; a non-collinear geometry generically gives PD_2P full rank $M - 1$ and every direction opens at order 2. Measured at dps $= 60$ over two independent p-ratios: collinear $M = 3$ gives orders $(2, 4)$ and $M = 4$ gives $(2, 4, 6)$, against $(2, 2)$ for the equilateral triangle, $(2, 2, 2)$ for the tetrahedron and $(2, 2)$ for a bent water-like triangle, with rank$(PD_2P) = 1, 1, 2, 3, 2$ respectively. This is the cluster-size phenomenon of Batenkov, Demanet, Goldman and Yomdin [42], whose exponent is controlled by the largest cluster and whose $\ell = 2$ case is the two-center problem here. Two consequences, and the second is a limit on what is claimed below. Water's A_1 block is *bent*, hence in the full-rank class — which is why one tri$(1, 2, 1)$ on the rotated component suffices and the table below works. A *linear* polyatomic is not: matching a zero of order $2k$ requires a polynomial with a zero of the same order, so the single tri$(1, 2, 1)$ cures only the one order-2 direction there. The lever is therefore established for $M = 2$ and for non-collinear $M = 3$; the collinear case is open and is not claimed.

For water's A_1 block, $A(\chi) = \left[\begin{smallmatrix} 1 & \sqrt{2}a_{\text{OH}} \\ \sqrt{2}a_{\text{OH}} & 1 + a_{\text{HH}} \end{smallmatrix}\right]$ degenerates to $\left[\begin{smallmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{smallmatrix}\right]$, singular with null direction $v \propto (\sqrt{2}, -1)$ and trace 3. Rotating the two-dimensional block space by v and applying tri$(1, 2, 1)$ to that component alone:

N	cond(A_1)	preconditioned
12	183.0	38.45
48	2696.1	43.62
192	41699.7	**44.06**

The raw column reproduces the $N^{1.97}$ of the probe above independently ($N^{1.96}$ here); the preconditioned column

is bounded, with increments collapsing geometrically $(3.94, 1.24, 0.35, 0.09)$. The constant is larger than the diatomic 2.23, but the growth—which is what makes the metric penalty basis-dependent—is gone. Two controls separate what is doing the work, and the first cannot do the job an earlier version of this paragraph gave it. Uniform banding, blockdiag(P, P), is $I_2 \otimes \mathrm{tri}(1, 2, 1)$ while the rotation is $V \otimes I$: the two *commute* (to 3×10^{-13} at $N = 192$), so that control returns the identical spectrum in the rotated and unrotated frames and is blind to the rotation by construction. What it does establish is that banding *alone* is not the lever — $194 \to 42008$ over the same range (the raw table's $N = 12$–192), an exponent of $N^{1.94}$, indistinguishable from the raw column's $N^{1.96}$ on this grid, i.e. no material change. (The interior $N = 48$ uniform value is 2766; an earlier draft printed it as the low endpoint, which reproduced a spurious $N^{0.98}$.) The discriminating control is the *selective* preconditioner blockdiag(P, I) applied in the *unrotated* frame, and it fails in the strong direction: conditioning grows as $N^{3.79}$ and reaches 4.4×10^6 at $N = 192$, $106\times$ *worse* than leaving the metric untreated. Alignment to the null direction is therefore load-bearing rather than incidental: the same band is worse than nothing in the wrong frame and flat at 44 in the right one.

[RESOURCE MODEL] Priced end to end the lever is smaller than the conditioning table suggests, and saying so precisely is more useful than the raw ratio. Two costs must be tracked rather than one: the QSVT degree d_{inv}, and the *subnormalization* α of the block-encodings, which multiplies when encodings are composed.

The amplitude is not negotiable. Any X with $X^\dagger S X = I$ satisfies $X = S^{-1/2}U$ for some unitary U, so

$$\|X\| = \|S^{-1/2}\| \quad \text{exactly, independently of how } X \text{ is factored} \tag{15}$$

(verified to 6×10^{-13} over $n = 20$–160). *No factorization can lower the amplitude floor*, and the untreated route already attains it. What preconditioning buys is depth, and only depth. The factor $P^{-1/2}$ is exactly implementable and costs no calls to any encoding of the metric: the DST-I has an $O(\log^2 N)$ quantum circuit [44], and the diagonal $\lambda_k^{-1/2} = \big(2{+}2\cos\frac{k\pi}{n+1}\big)^{-1/2}$ is computed from the index.

route	α	d_{inv}	$\alpha\, d_{\mathrm{inv}}$
untreated	125.4	3.1×10^5	3.9×10^7
preconditioned, G composed	3196	16.1	5.1×10^4
preconditioned, G direct	125.5	24.3	3.1×10^3

(at $n = 160$, $kR = 2$, $\epsilon_E = 1.6$ mHa). In exponents, which is the honest form: untreated $\alpha \sim n$ and $d_{\mathrm{inv}} \sim n^2$, so the product scales as n^3; preconditioned with G obtained by *composing* $P^{-1/2}$ with $I - C$, the degree goes flat but α inherits $\|P^{-1/2}\|^2 \sim n^2$, giving n^2; with a direct encoding of G at the floor it would be n. The third row is not hypothetical: the direct encoding exists, and the rest of this subsection constructs it. **So the lever is worth two powers of n, taking the metric penalty from n^3 to n.**

[SYMBOLIC + MEASURED] The construction is available because G's symbol is a *ratio* of two symbols that vanish to the same order. Both $1 - \sigma$ and g have quadratic zeros at $\chi = \pi$, so the quotient tends to $(kR)^2/24$ there, and at $\chi \to 0$ it tends to $\frac{1}{4}$; in the variable $s = kR\cot(\chi/2)$,

$$\mathrm{ratio}(s) = \big(1 - j_0(s)\big)\frac{s^2 + (kR)^2}{4s^2}, \qquad \|\mathrm{ratio}\|_\infty = 0.3716 = \|G\| \tag{16}$$

—the sup coincides with $\|G\|$, so a circulant-embedded encoding of the Toeplitz-minus-Hankel matrix B built from this symbol's own cosine coefficients carries $\alpha = 0.372$ rather than the composed 3196. That recovers exactly the factor the composition wasted.

B is *not* G: they differ by the finite-section commutator, measured at 11.5–11.8% in operator norm and *not growing* with n. The point is that the difference does not matter, because B enters only as a whitening. Any X with $X^\dagger S X = I$ preserves the spectrum (Eq. (15)), and taking $X = P^{-1/2}B^{-1/2}$ gives $X^\dagger(I - C)X = B^{-1/2}GB^{-1/2}$, whose conditioning is *bounded*—$1.222, 1.228, 1.232, 1.234$ at $n = 20, 40, 80, 160$, settling near 1.23—so one further $O(1)$-degree transformation absorbs it. And $\|X\|$ lands on the amplitude floor: $\|P^{-1/2}B^{-1/2}\|/\|(I - C)^{-1/2}\| = 1.0097, 1.0049, 1.0025, 1.0013$ over the same range, tightening toward 1.

n	cond(B)	$\|G - B\|/\|G\|$	cond$(B^{-1/2}GB^{-1/2})$	$\|X\|$/floor
20	2.198	0.115	1.222	1.0097
80	2.228	0.118	1.232	1.0025
160	2.229	0.118	**1.234**	**1.0013**

So the whole metric factor is three explicitly-known pieces: a DST-I, a diagonal computed from the index, and a degree-$\approx$ 24 QSVT on a directly-encoded Toeplitz-minus-Hankel matrix whose subnormalization is $O(1)$—none of it growing with basis size (`tests/test_paper60_direct_encoding.py`).

Scope. What is established is the *conditioning* statement, on s-sector shared-scale bases at $M = 2$ and *non-collinear* $M = 3$ (the collinear case is open and is not claimed), together with the pricing above under Table I's cost model, and the direct construction of Eq. (16). What is *not* established is the circuit: the $O(\log^2 N)$ sine transform is cited rather than compiled, and the circulant embedding of B is standard but not laid out here, so the counts remain a resource model rather than a gate count. And by the block-structure result above, none of this recovers ℓ-selection, which no congruence can—the metric is now cheap to *apply*, and the sparsity it destroys stays destroyed.

[MEASURED] The large-R lever is quantified alongside: cond(S) falls to 2.2 at $R = 10$ bohr (at the sweep's

$N = 12$, $d_{\rm inv}$: $862 \to 17$ over $R = 1.4 \to 10$; at Table I's $N = 16$ the $R = 1.4$ figure is larger still, so the lever is stronger than these numbers suggest), while the Gaussian metric stays $\sim 2 \times 10^3$—its intra-center linear dependence does not relax with separation.

[MEASURED] Two probes price the derived law's exploitability (resource-model tier; CHANGELOG v4.105.0). *Spectrum-aware inversion*—replacing the worst-case interval approximation for $S^{-1/2}$ by constrained interpolation on the *known* spectrum $\{1 \pm \sigma_k\}$ (classically computable by an $O(N^3)$ values-only SVD that touches no integral sparsity; the asymptotic law alone is 11–21% off at $\lambda_{\min}$ and does not suffice)—buys a constant factor $\approx$ 8–10$\times$ at the *same* κ exponent, and the $\Theta(\kappa)$ floor of Theorem 73 of [13] for *both* problems bounds it there (an argument, not a measurement: the floors hold for the interpolation and interval families alike, because the small eigenvalues are spaced at exactly the scale on which $x^{-1/2}$ varies). At this table's $N = 16$ point the honest chain is model 1050 $\to$ same-convention minimax 209 $\to$ aware 27. An *exponent-changing* variant exists—a PSD shift of $[\lambda_{\min}, \lambda_{\max}]$ onto $[-1, 1]$, needing only Eq. (13)'s closed-form endpoints, gives degree $\approx 2.8\sqrt{\kappa}$—but is gated by its LCU subnormalization (the naive 1-norm ≈ 3 squeezes the spectrum back to linear-in-κ, *worse* than baseline; with $\sim 3\times$ amplification the net at $\kappa = 864$ is $\approx 5 \times 10^2$ effective queries vs $\approx 1.9 \times 10^3$)—scoped, not priced beyond this. Separately, an *analytic momentum-space factorization* of the ERI tensor (the closed-form momentum densities as a continuous factorization index) matches but does not beat plain density fitting: eigen-Cholesky DF on the same tensor reaches within 13% of its 1-norm at rank $\le n_{\rm orb}^2$, versus 2.5–4.3 $\times 10^4$ momentum leaves—the observed gains are DF's, not the momentum representation's.

B. Many-electron molecules: the metric does not compound

For a many-electron molecule the generalized-Sturmian secular equation is still $[T^0 + T' - p_\kappa \not\kappa]B = 0$—a *standard* eigenproblem, the $-p_\kappa \not\kappa$ being the identity, not a metric (Avery's general form [3]); only the atomic conveniences "T^0 diagonal" and "T' pure numbers" are lost. The resource-relevant question is whether the Shibuya–Wulfman metric penalty of Sec. VI *compounds* with electron number. It need not.

[SYMBOLIC + MEASURED] A k-electron configuration is a Slater determinant of one-electron molecular orbitals, so its configuration-space overlap is the k-th compound matrix [18–21] of the one-electron overlap, with condition number $(\lambda_1 \cdots \lambda_k)/(\lambda_{N-k+1} \cdots \lambda_N)$ in the one-electron overlap spectrum $\{\lambda_i\}$. Built from the *raw* L^2-normalized many-center Coulomb–Sturmians this *grows* with electron number—on the H_2-geometry two-center basis ($N = 6$), $1\times, 8\times, 9\times$ $\mathrm{cond}(S_{L^2})$ at $k = 1, 2, 3$ (worst at half-filling), bounded by $\mathrm{cond}(S_{L^2})^{\min(k,N-k)}$ and driven by the near-linear-dependence eigenvalue. Built instead from the *molecular Sturmians*—the Shibuya–Wulfman eigenvectors, $S_{\rm SW}$-orthonormal by construction—the configuration overlap is the *identity at every electron number*—an algebraic consequence of the compound-matrix identity, not a numerical finding; only the $1\times, 8\times, 9\times$ L^2 figures above are measured. The metric is therefore paid once, at the one-electron molecular-orbital construction ($\mathrm{cond}(S_{\rm SW})$ above, with the gerade lever), and the N-electron secular equation inherits an identity metric: **the molecular metric penalty is one-electron, not N-electron**. It is the on-device analogue of the once-executed classical mean-field step that lets production Gaussian simulation carry no metric into the many-body Hamiltonian.

[MEASURED] The compound-matrix identity also fixes *where* the two centers decompactify: because a k-electron configuration overlap is the k-th compound of the one-electron overlap, the many-electron decompactification front moves only through screening (the occupied decay length), while correlation moves a distinct signed cross-center coherence front through the antibonding occupation. Both fronts, with their measured H_2 and HeH^+ tests, are developed in Paper 58 [15] (Sec. "the continuous side," the many-electron front); they are bonding physics, not encoding cost, and ride in here only on the shared compound-matrix identity.

[MEASURED] The interelectron matrix T' is assembled from two-center electron-repulsion integrals, and these are in hand. We evaluate the shared-scale Coulomb–Sturmian two-center ERIs $(pq|rs)$ numerically (a single multipole route about one nucleus, so every center class is uniform) and validate them against the framework's *exact* closed form: the one-center $(1s\,1s|1s\,1s) = \frac{5}{8}$ to 4×10^{-6}, and the two-center $(AA|BB)$ to $\sim 10^{-4}$ against `aabb_value` at $R \in \{1.4, 2.0, 3.0\}$ (the closed forms themselves are weight-one, π-free $\{E_1, \ln, \gamma\}$ [15, 16]). With these, the interacting method binds a molecule: minimal-basis ($1s$ on each center, common scale) H_2 in the metric-free molecular-Sturmian basis has an interior minimum at $R \approx 1.6$ bohr with $E_{\rm tot} \approx -1.09$ Ha—the single-ζ minimal-basis value, underbinding the exact -1.174 Ha as s-only single-ζ must, above the exact energy throughout.

[OPEN, with the obstruction identified] What remains is the block-encoding 1-norm of the full N-electron secular matrix $T^0 + T'$, and here the molecular case parts from the atomic one at a structural level, not merely a computational one. The atomic sublinearity of Eq. (6) rides on the *clean diagonal* $T^0 = ZR_\nu$, which exists only because (i) the Goscinskian orbitals scale *with* p_κ (charge $Q_\nu = p_\kappa/R_\nu$), so the p_κ in $\langle 1/r \rangle$ cancels the $-p_\kappa^{-1}$ prefactor and M is a p_κ-independent linear matrix, and (ii) on a *single* center $\langle 1/r \rangle = Q/n^2$ collapses the collective scale to the root-sum-of-squares $T^0_{\rm bare} = \sqrt{\sum_p k_p^2} = ZR_\nu$ (we verify: for He $1s^2$, $\sqrt{k^2 + k^2} = 2\sqrt{2} = ZR_\nu$ gives the exact bare -4 Ha; a naive sum of

scales gives -8). On *two* centers the nuclear attraction $\langle 1/r_A\rangle + \langle 1/r_B\rangle$ acquires off-center terms depending on $Q_\nu R$, so T^0 is neither diagonal nor p_κ-independent, and a fixed-scale molecular-Sturmian shortcut is not the isoenergetic matrix at all—it fails the non-interacting limit (returning $p_\kappa = 2k_{\sigma_g}$ where the correct value is $\sqrt{2}\,k_{\sigma_g}$). A faithful molecular 1-norm therefore requires the p_κ-scaled molecular Goscinskian (config charge $Q_\nu = p_\kappa/R_\nu$, two-center nuclear T^0 and two-center ERIs at the config scale, solved self-consistently; validated at $R\to\infty = 2$ H atoms and $R\to 0 =$ He). The obstruction is thus not that the integrals are hard—they close (Papers [15, 16])—but that the single-center diagonal-T^0 structure that *produced* the sublinearity does not transfer.

[MEASURED] We can nonetheless price the practical alternative. Building the two-center mixed-charge integrals (overlap, nuclear attraction, and ERIs, validated against the exact closed forms and against $\langle 1s|1/r_B|1s\rangle$ to $\sim 10^{-4}$) and a full two-electron CI in the Loewdin-orthonormalized Sturmian basis yields a *validated* molecular solver—H_2 dissociates correctly to two H atoms ($E_{\rm tot}\to -1.0$ Ha at $R = 6$) and binds above the exact -1.174 Ha at $R_{\rm eq}$. Its *standard* block-encoding 1-norm $\lambda = \sum_{pq}|h_{pq}| + \sum_{pqrs}|(pq|rs)|$ grows polynomially, $\lambda \sim n_{\rm orb}^{2.2}$, with no sublinear behavior—the atomic advantage of Eq. (6) is a property of the isoenergetic secular matrix, not of the basis, and it does not survive the loss of the diagonal T^0. For molecules the resource question therefore answers in the ordinary way: the block-encoding cost scales polynomially, and the metric levers of Secs. VI A–VI B (the gerade sector, large-R, one-electron-only metric) are the real savings, not a sublinear matrix.

VII. CONCLUSION AND SCOPE

The message is a sharp split. *For atoms*, the isoenergetic generalized-Sturmian reformulation converts electronic structure into a standard, metric-free eigenproblem—a clean, apparently novel quantum secular equation in which the two feared cost risks of the genre are structurally absent. Its block-encoding 1-norm grows more slowly than the matrix dimension over every basis we can compute, and increasingly so. But the sharper statement is a trade-off rather than an efficiency: that growth belongs to the fixed-$l_{\rm max}$ basis-growth rule, reverses under full-shell growth, and is paid for by an accuracy floor whose mechanism is Eq. (10): the metric-free form exists only at the scale $\lambda = p_\kappa$, and that scale is not the variational optimum. The floor is a *ground-state* pathology — 6.8 mHa there at $K = 452$, but 1.72 mHa for $2\,^1S$ at identical encoding cost, and—on the s-only ladder at $K = 105$—falling by 4.3×, 3.0× and 2.6× across the first three rungs of the ^1S ladder. Read as a ground-state route to chemical accuracy it is not one; read as a cheap fixed-accuracy tier, and especially as an *excited-state* method — which is where the generalized-Sturmian literature has always been strongest — it is a real operating point. The open question we leave is whether a weighting potential of different radial shape can lower the ground-state floor without breaking Eq. (9). *For molecules*, the Shibuya–Wulfman metric returns; it is genuinely better-conditioned than the L^2 overlap—its intra-center block is the identity—and much better at large separation, but its inter-center coupling still costs conditioning that grows with basis size. Two qualifications lift this from a caveat to a payoff. The first is symmetry: because the H_2^+ ground state is gerade and the gerade sector's conditioning is *flat* in basis size (at the constant 2.555041 . . ., so flat rather than absent—Table I still charges it), the ground-state block-encoding pays a metric penalty that does not grow, and the growth is confined to the excited/ungerade states. That first lever is an *equivalent*-center property, and a probe on water (C_{2v}) shows a symmetry-unique heavy center reinstates the raw growth in the ground-state (A_1) block (Sec. VI). The second qualification is what removes that limit: because the ill-conditioning is a symbol zero of known order and location, a band-Toeplitz preconditioner bounds it flat in basis size without requiring equivalent centers, and it reaches water's A_1 block (Sec. VI A), provided the band is aligned to the symbol's null direction — the same band applied in the unrotated frame is worse than leaving the metric untreated. Established on s-sector shared-scale bases at $M = 2$ and non-collinear $M = 3$; the collinear case is open and is not claimed. And the penalty does not compound with electron number: the Shibuya–Wulfman metric is discharged once, at the one-electron molecular-orbital construction, after which the many-electron secular equation carries an identity metric (Sec. VI B)—the metric cost is one-electron, not N-electron. The atomic case is a result; the molecular case is a frontier that is *removable on the conditioning axis* within that scope, capped on locality by the symbol's other pole, and structurally closed on ℓ-sparsity.

We have not run the algorithm on hardware, and we do not claim a resource advantage over production quantum chemistry: the systems where the encoding is demonstrably cheap are small. What we claim is structural and measured—that the generalized-Sturmian method's classical elegance (no self-consistency, energies direct, pure-number matrices) is also, for atoms, a near-ideal quantum encoding—and that this reframes the on-device-generation intuition from a hope into a characterized boundary, clean on one side and open on the other.

ACKNOWLEDGMENTS

The generalized-Sturmian formulation used here is that of J. E. Avery and J. S. Avery, as are the Shibuya–Wulfman integrals and the split-shell account of why the helium ground state in particular resists the method. To these this paper adds its reading as a quantum-encoding target, the resource measurements, and the results we

claim as our own, which are (the list is enumerated, not exhaustive of the paper's smaller observations): the scale-lock identity Eq. (10) together with the one-body Coulomb metric Eq. (9) it rests on; the interlacing bound Eq. (5); the identification of the Shibuya–Wulfman matrix *as* the V_0-weighted overlap in Eq. (11); and the variational bound for the fixed-scale metric-free problem. Developed with an AI-augmented workflow under principal-investigator direction, with the headline numerical claims validated against analytical benchmarks and a regression test suite.

Appendix A: Radial-domain convergence rule

[MEASURED] The Goscinskian configuration $\nu = (l, n_a, n_b)$ carries weighted charge $Q_\nu = p_\kappa/R_\nu$ with $R_\nu = \sqrt{n_a^{-2} + n_b^{-2}}$, and its orbital n is hydrogenic at exponent $a = Q_\nu/n$. Its radial density therefore peaks at $r_{\rm pk} = n/a = n^2 R_\nu$, and the mismatched pairs $(l, 1, n)$ at $Q_\nu \approx 1$ reach furthest. A radial domain must satisfy

$$R_{\max} \;\geq\; 3 \max_{\nu} \max_{j} n_j^2 R_\nu \;\approx\; 3\, n_{\max}^2 \qquad \text{(A1)}$$

for the 1-norm exponents to be converged. Measured relative error of $\|T'\|_1^{\rm off}$ against a $R_{\max} = 5n_{\max}^2$ reference: a fixed 60-bohr domain gives 1.1×10^{-2} at $K = 74$ rising to 2.1×10^{-1} at $K = 340$; $c = 1$ gives $\sim 10^{-2}$ *falling* with K; $c = 2$ gives $\sim 10^{-4}$; $c = 3$ gives 2.15–2.16×10^{-7}, converged. Two independent quadrature rules were run to convergence: a uniform mesh at 1.5×10^6 points and a graded $r = Rt^2$ mesh at 2.4×10^5. On the relative form quoted here they read 2.147 and 2.156×10^{-7} (a spread of 0.43%); on the *absolute* box truncation, measured against a well-resolved $c = 12$ reference, they read 2.161 and 2.163×10^{-7} (0.08%). An earlier version of this sentence gave only the order of magnitude. Exponents are converged to ± 0.001 at $c = 2$ and $\pm 10^{-4}$ at $c = 3$.

The distinction that matters is not the size of the error but its *sign of drift*. A domain that scales with the basis produces an error that shrinks with K and biases a log–log slope negligibly; a fixed domain produces an error that *grows* with K, i.e. with the fit variable, and biases the slope upward. The mechanism is normalization: an orbital truncated at 60 bohr retains as little as 2.7% of its norm at $n = 14$ and is then renormalized to unity, manufacturing a compact pseudo-orbital whose repulsion integral is $O(1)$ where the true coupling decays as n^{-2}. Equal-n configurations and $1s^2$ are unaffected, so single-configuration checks — including $E(1s^2) = -729/256$ and the $(5/8)Q$ unit-charge integral — are blind to it.

[1] J. Avery, *Hyperspherical Harmonics: Applications in Quantum Theory* (Kluwer Academic, Dordrecht, 1989).

[2] J. E. Avery and J. S. Avery, *Generalized Sturmians and Atomic Spectra* (World Scientific, Singapore, 2006).

[3] J. E. Avery, *New Computational Methods in the Quantum Theory of Nano-structures*, Ph.D. thesis, University of Copenhagen (DIKU), 2011.

[4] J. E. Avery, *The Generalized Sturmian Method: Development, Implementation and Applications in Atomic Physics*, M.Sc. thesis, University of Copenhagen, 2008.

[5] M. F. Herbst, J. E. Avery, and A. Dreuw, "Quantum chemistry with Coulomb Sturmians: Construction and convergence of Coulomb Sturmian basis sets at Hartree–Fock level," Phys. Rev. A **99**, 012512 (2019); arXiv:1811.05777.

[6] D. Calderini, S. Cavalli, C. Coletti, G. Grossi, and V. Aquilanti, "Hydrogenoid orbitals revisited: From Slater orbitals to Coulomb Sturmians," J. Chem. Sci. **124**, 187 (2012).

[7] R. Babbush, N. Wiebe, J. McClean, J. McClain, H. Neven, and G. K.-L. Chan, "Low-depth quantum simulation of materials," Phys. Rev. X **8**, 011044 (2018); arXiv:1706.00023.

[8] Y. Su, D. W. Berry, N. Wiebe, N. Rubin, and R. Babbush, "Fault-tolerant quantum simulations of chemistry in first quantization," PRX Quantum **2**, 040332 (2021); arXiv:2105.12767.

[9] G. H. Low and I. L. Chuang, "Hamiltonian simulation by qubitization," Quantum **3**, 163 (2019); arXiv:1610.06546.

[10] J.-M. Liang, S.-Q. Shen, M. Li, and S.-M. Fei, "Quantum algorithms for the generalized eigenvalue problem," Quantum Inf. Process. **21**, 23 (2022); arXiv:2112.02554.

[11] G. Rajchel-Mieldzioć, S. Pliś, and E. Zak, "Quantum algorithm for solving generalized eigenvalue problems with application to the Schrödinger equation," arXiv:2506.13534 (2025).

[12] U. Baek *et al.*, "Say NO to optimization: A non-orthogonal quantum eigensolver," PRX Quantum **4**, 030307 (2023); arXiv:2205.09039.

[13] A. Gilyén, Y. Su, G. H. Low, and N. Wiebe, "Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics," in *Proc. 51st ACM STOC* (2019), p. 193; arXiv:1806.01838. (Theorem and corollary numbering here follows the arXiv version; the STOC proceedings text is the 12-page abridgement.)

[14] A. M. Childs, R. Kothari, and R. D. Somma, "Quantum algorithm for systems of linear equations with exponentially improved dependence on precision," SIAM J. Comput. **46**, 1920 (2017); arXiv:1511.02306.

[15] J. Loutey, "Angular Sparsity Is an Atomic-Sector Property: The Abelian Residue of Multi-Center Bonding" (Paper 58, Geometric Vacuum series).

[16] J. Loutey, "The Three-Center Electron-Repulsion Integral Is an Elliptic Bessel Moment" (Paper 59, Geometric Vacuum series).

[17] A. T. Amos and G. G. Hall, "Single determinant wave functions," *Proc. R. Soc. Lond. A* **263**, 483 (1961).

[18] H. F. King, R. E. Stanton, H. Kim, R. E. Wyatt, and R. G. Parr, "Corresponding orbitals and the nonorthogonality problem in molecular quantum mechanics," *J. Chem. Phys.* **47**, 1936 (1967).

[19] P.-O. Löwdin, "Quantum theory of many-particle systems. I. Physical interpretations by means of density matrices, natural spin-orbitals, and convergence problems in the method of configurational interaction," *Phys. Rev.* **97**, 1474 (1955).

[20] F. Prosser and S. Hagstrom, "On the rapid computation of matrix elements," *Int. J. Quantum Chem.* **2**, 89 (1968).

[21] H. G. A. Burton, "Generalized nonorthogonal matrix elements: Unifying Wick's theorem and the Slater–Condon rules," *J. Chem. Phys.* **154**, 144109 (2021).

[22] E. Koridon, S. Yalouz, B. Senjean, F. Buda, T. E. O'Brien, and L. Visscher, "Orbital transformations to reduce the 1-norm of the electronic structure Hamiltonian for quantum computing applications," *Phys. Rev. Research* **3**, 033127 (2021).

[23] R. Babbush, D. W. Berry, J. R. McClean, and H. Neven, "Quantum simulation of chemistry with sublinear scaling in basis size," *npj Quantum Inf.* **5**, 92 (2019).

[24] C. Jordan, "Essai sur la géométrie à n dimensions," *Bull. Soc. Math. France* **3**, 103 (1875).

[25] Å. Björck and G. H. Golub, "Numerical methods for computing angles between linear subspaces," *Math. Comp.* **27**, 579 (1973).

[26] V. Eijkhout and P. Vassilevski, "The role of the strengthened Cauchy–Buniakowskii–Schwarz inequality in multilevel methods," *SIAM Rev.* **33**, 405 (1991).

[27] P. Hartman and A. Wintner, "The spectra of Toeplitz's matrices," *Amer. J. Math.* **76**, 867 (1954).

[28] P.-O. Löwdin, "On the non-orthogonality problem connected with the use of atomic wave functions in the theory of molecules and crystals," *J. Chem. Phys.* **18**, 365 (1950).

[29] J. C. Slater and G. F. Koster, "Simplified LCAO method for the periodic potential problem," *Phys. Rev.* **94**, 1498 (1954).

[30] T. A. Rokob, Á. Szabados, and P. R. Surján, "A note on the symmetry properties of Löwdin's orthogonalization schemes," *Collect. Czech. Chem. Commun.* **73**, 937 (2008) (Zahradník Festschrift issue).

[31] S. Jaffard, "Propriétés des matrices bien localisées près de leur diagonale et quelques applications," *Ann. Inst. H. Poincaré Anal. Non Linéaire* **7**, 461 (1990).

[32] K. Gröchenig and M. Leinert, "Symmetry and inverse-closedness of matrix algebras and functional calculus for infinite matrices," *Trans. Amer. Math. Soc.* **358**, 2695 (2006).

[33] T. A. Driscoll and B. Fornberg, "Interpolation in the limit of increasingly flat radial basis functions," *Comput. Math. Appl.* **43**, 413 (2002).

[34] NIST Digital Library of Mathematical Functions, `https://dlmf.nist.gov/`, F. W. J. Olver *et al.*, eds.

[35] T. A. Loring, "Principal angles and approximation for quaternionic projections," *Ann. Funct. Anal.* **5**(2), 176 (2014); arXiv:1306.1923.

[36] H. J. Monkhorst and B. Jeziorski, "No linear dependence and multi-center integral problems in momentum space quantum chemistry," *J. Chem. Phys.* **71**(12), 5268–5269 (1979); doi:10.1063/1.438337.

[37] C. E. Wulfman and Y. Takahata, "Noninvariance groups in molecular quantum mechanics. I," *J. Chem. Phys.* **47**(2), 488–498 (1967); doi:10.1063/1.1711921.

[38] E. Red and C. A. Weatherford, "Derivation of a general formula for the Shibuya–Wulfman matrix," *Int. J. Quantum Chem.* **100**, 208–213 (2004); doi:10.1002/qua.20122.

[39] O. Goscinski, "Conjugate eigenvalue problems and generalized Sturmians," *Adv. Quantum Chem.* **41**, 51–56 (2002); doi:10.1016/S0065-3276(02)41046-5. A presentation of the previously unpublished *Conjugate Eigenvalue Problems and the Theory of Upper and Lower Bounds*, Preliminary Research Report No. 217, Quantum Chemistry Group, Uppsala University (1968).

[40] T. I. Shibuya and C. E. Wulfman, "Molecular orbitals in momentum space," *Proc. Roy. Soc. A* **286**, 376 (1965).

[41] F. Ahmad, E. S. Al-Aidarous, D. A. Alrehaili, S.-E. Ekström, I. Furci and S. Serra-Capizzano, "Are the eigenvalues of preconditioned banded symmetric Toeplitz matrices known in almost closed form?," *Numer. Algorithms* **78**(3), 867–893 (2018); doi:10.1007/s11075-017-0404-z.

[42] D. Batenkov, L. Demanet, G. Goldman and Y. Yomdin, "Conditioning of partial nonuniform Fourier matrices with clustered nodes," arXiv:1809.00658 (2018; rev. 2019).

[43] S. Barthelmé and K. Usevich, "Spectral properties of kernel matrices in the flat limit," *SIAM J. Matrix Anal. Appl.* **42**(1), 17 (2021); arXiv:1910.14067.

[44] A. Klappenecker and M. Rötteler, "Discrete cosine transforms on quantum computers," in *Proc. IEEE R8-EURASIP Symp. Image and Signal Processing and Analysis (ISPA)*, Pula, Croatia, pp. 464–468 (2001); arXiv:quant-ph/0111038.

[45] S. Serra, "Optimal, quasi-optimal and superlinear band-Toeplitz preconditioners for asymptotically ill-conditioned positive definite Toeplitz systems," *Math. Comp.* **66**, 651 (1997).

[46] M. Kac, W. L. Murdock, and G. Szegő, "On the eigenvalues of certain Hermitian forms," *J. Rational Mech. Anal.* **2**, 767 (1953).

[47] A. Böttcher and H. Widom, "From Toeplitz eigenvalues through Green's kernels to higher-order Wirtinger–Sobolev inequalities," in *The Extended Field of Operator Theory*, Operator Theory: Advances and Applications, Birkhäuser, Basel (2007), pp. 73–87; doi:10.1007/978-3-7643-7980-3_4; arXiv:math/0412269 (2004).

[48] P. R. Halmos, "Two subspaces," *Trans. Amer. Math. Soc.* **144**, 381 (1969).

[49] A. Böttcher and I. M. Spitkovsky, "A gentle guide to the basics of two projections theory," *Linear Algebra Appl.* **432**, 1412 (2010).